

SHARP DECAY ESTIMATES FOR GLOBAL SOLUTIONS TO THE INCOMPRESSIBLE ROTATING NAVIER–STOKES EQUATIONS

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ABSTRACT. In this paper, we consider the three-dimensional incompressible rotating Navier–Stokes equations and establish the sharp L^p decay estimates of global solutions. We reveal that the optimal L^p decay rates for $2 < p < \infty$ are strictly faster than those obtained in existing results by interpolation between the L^2 unitary identity and L^∞ dispersive estimates, although the known endpoint cases are sharp. Moreover, the optimality of decay rates is also proved by choosing a class of initial data. The underlying mechanism lies in the anisotropic degeneracy of the oscillatory integrals arising from the Coriolis force.

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1. INTRODUCTION

In this paper, we consider the initial value problem for 3D incompressible rotating Navier–Stokes equations:

$$\begin{cases} \partial_t u - \Delta u + \Omega e_3 \times u + (u \cdot \nabla)u + \nabla P = 0, & t > 0, x \in \mathbb{R}^3, \\ \operatorname{div} u = 0, & t \geq 0, x \in \mathbb{R}^3, \\ u(0, x) = u_0(x), & x \in \mathbb{R}^3. \end{cases} \quad (1.1)$$

Here, $u = u(t, x) : [0, \infty) \times \mathbb{R}^3 \rightarrow \mathbb{R}^3$ and $P = P(t, x) : (0, \infty) \times \mathbb{R}^3 \rightarrow \mathbb{R}$ stand for the unknown velocity and pressure of the fluid, respectively, whereas $u_0 = u_0(x) : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is the given initial velocity satisfying the compatibility condition $\operatorname{div} u_0 = 0$. For the Coriolis term $\Omega e_3 \times u$, the constant $\Omega \in \mathbb{R}$ represents the angular speed of the rotation around the vertical direction $e_3 = (0, 0, 1)$. The mathematical analysis of (1.1) was initiated by Babin–Mahalov–Nicolauenko [2–4], and has subsequently been developed in a large body of work. For the global well-posedness, we refer the readers to [7–9, 12, 19, 20, 22]; for the fast rotation limit, to [7–9, 13, 26]; and for the large time behavior, to [1, 10, 21, 29].

Applying the Helmholtz projection $\mathbb{P} = I + \nabla(-\Delta)^{-1} \operatorname{div}$ to the first equation of (1.1) and using the Duhamel principle, we may rewrite (1.1) as the integral equation

$$u(t) = \mathcal{T}_\Omega(t)u_0 - \int_0^t \mathcal{T}_\Omega(t - \tau) \mathbb{P} \operatorname{div}(u(\tau) \otimes u(\tau)) d\tau,$$

where $\{\mathcal{T}_\Omega(t)\}_{t \geq 0}$ is the semigroup generated by the Stokes–Coriolis operator $-\Delta + \Omega \mathbb{P} e_3 \times$. From the work of Hieber–Shibata [17], it is known that this semigroup is represented

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explicitly as

$$\begin{aligned}\mathcal{T}_\Omega(t)u_0 &:= e^{t(\Delta - \Omega \mathbb{P}e_3 \times \mathbb{P})}u_0 \\ &= e^{t\Delta} e^{it\Omega \frac{D_3}{|D|}} P_+(D)u_0 + e^{t\Delta} e^{-it\Omega \frac{D_3}{|D|}} P_-(D)u_0.\end{aligned}$$

where the Fourier multiplier $P_\pm(D) = \mathcal{F}^{-1}P_\pm(\xi)\mathcal{F}$ is defined via

$$P_\pm(\xi) = \frac{1}{2}(I \pm iR(\xi)), \quad R(\xi) := \frac{1}{|\xi|} \begin{pmatrix} 0 & -\xi_3 & \xi_2 \\ \xi_3 & 0 & -\xi_1 \\ -\xi_2 & \xi_1 & 0 \end{pmatrix}. \quad (1.2)$$

One of the key features distinguishing the rotating Navier–Stokes equations from the classical Navier–Stokes equations is that their linearized semigroup includes the dispersive evolution group $\{e^{i\tau \frac{D_3}{|D|}}\}_{\tau \in \mathbb{R}}$ generated by the Coriolis force. To see the dispersive nature, let us focus on the Fourier integral

$$e^{i\tau \frac{D_3}{|D|}} \varphi(x) = \frac{1}{(2\pi)^3} \int_{\mathbb{R}^3} e^{ix \cdot \xi} e^{i\tau \frac{\xi_3}{|\xi|}} \widehat{\varphi}(\xi) d\xi,$$

where $\varphi \in \mathcal{S}(\mathbb{R}^3)$ is a given frequency amplitude satisfying $\text{supp } \widehat{\varphi} \subset \{\xi \in \mathbb{R}^3; 1/4 \leq |\xi| \leq 4\}$. In [22, 23], the following dispersive decay estimate was shown:

$$\left\| e^{i\tau \frac{D_3}{|D|}} \varphi \right\|_{L^\infty} \leq C(1 + |\tau|)^{-1}, \quad \tau \in \mathbb{R}, \quad (1.3)$$

with some positive constant C depending only on φ . We emphasize that, due to the classical Littman lemma, the decay rate -1 comes from the following degeneracy of the phase function

$$\text{rank } \nabla^2 \left(\frac{\xi_3}{|\xi|} \right) \geq 2 \quad \text{on } \text{supp } \widehat{\varphi}.$$

We also remark that Koh–Lee–Takada [23] proved the polynomial decay order -1 is sharp. On the other hand, the Plancherel theorem immediately implies

$$\left\| e^{i\tau \frac{D_3}{|D|}} \varphi \right\|_{L^2} = \|\varphi\|_{L^2}, \quad \tau \in \mathbb{R}. \quad (1.4)$$

Then, by employing the interpolation between (1.3) and (1.4), the L^p decay rates of the group $e^{i\tau \frac{D_3}{|D|}} \varphi$ is given by

$$\left\| e^{i\tau \frac{D_3}{|D|}} \varphi \right\|_{L^p} \leq C(1 + |\tau|)^{-(1-\frac{2}{p})}, \quad \tau \in \mathbb{R}, \quad 2 \leq p \leq \infty. \quad (1.5)$$

Taking the dispersive effect into account of the large time analysis for rotating Navier–Stokes flow, the linear L^1 – L^p type decay estimate was deduced by [10] as follows:

$$\left\| \nabla^m e^{t\Delta} e^{i\tau \frac{D_3}{|D|}} f \right\|_{L^p} \leq C t^{-\frac{3}{2}(1-\frac{1}{p})-\frac{m}{2}} (1 + |\tau|)^{-(1-\frac{2}{p})} \|f\|_{L^1}, \quad t > 0, \quad \tau \in \mathbb{R} \quad (1.6)$$

for $2 \leq p \leq \infty$ and $m \in \mathbb{N} \cup \{0\}$. Furthermore, it was proved by [10, 29] that the nonlinear solution u with the initial data in the class $u_0 \in L^1(\mathbb{R}^3) \cap \dot{H}^s$ ($1/2 < s < 9/10$) satisfies for every $2 \leq p \leq \infty$ and $m \in \mathbb{N} \cup \{0\}$ that

$$\|\nabla^m u(t)\|_{L^p} = o\left(t^{-\frac{3}{2}(1-\frac{1}{p})-\frac{m}{2}} (1 + |\Omega|t)^{-(1-\frac{2}{p})}\right)$$

as $t \rightarrow \infty$ and $1/2$ -enhanced decay estimates

$$\|\nabla^m u(t)\|_{L^p} \leq C t^{-\frac{3}{2}(1-\frac{1}{p})-\frac{m+1}{2}} (1 + |\Omega|t)^{-(1-\frac{2}{p})}, \quad t > 0,$$

provided that the additional condition $|x|u_0(x) \in L^1(\mathbb{R}^3)$ is imposed. Observe that the asymptotic expansion of Fujigaki–Miyakawa [11] type for some $2 < p < 3$ was established

in [10]. We refer to [1, 21] for the temporal decay estimates of solutions to the rotating fractional Navier–Stokes equations and rotating MHD equations.

However, it remains an open problem whether the additional decay rate $(1 + |\Omega|t)^{-(1-\frac{2}{p})}$ induced by the dispersive effects of the Coriolis force is optimal except for the endpoint cases $p = 2, \infty$. Let us stress that the optimality when $p = 2$ is classical in incompressible Navier–Stokes equations and the optimality when $p = \infty$ was verified in Koh–Lee–Takada [23]. *In this paper, we reveal that the dispersive decay rates $(1 + |\Omega|t)^{-(1-\frac{2}{p})}$ obtained by interpolation are in fact not optimal, and that the true optimal decay rates are strictly faster for $2 < p < \infty$.* More precisely, we prove in Section 3 that for $m \in \mathbb{N} \cup \{0\}$,

$$\left\| \nabla^m e^{t\Delta} e^{i\tau \frac{D_3}{|D|}} f \right\|_{L^p} \leq \begin{cases} Ct^{-\frac{3}{2}(1-\frac{1}{p})-\frac{m}{2}} (1 + |\tau|)^{-\frac{3}{2}(1-\frac{2}{p})} \|f\|_{L^1} & (2 \leq p < 4), \\ Ct^{-\frac{3}{2}(1-\frac{1}{p})-\frac{m}{2}} (1 + |\tau|)^{-(1-\frac{1}{p})} \|f\|_{L^1} & (4 < p \leq \infty) \end{cases}$$

for $f \in L^1(\mathbb{R}^3)$, which is better than (1.6). Moreover, for the case of $p = 4$, we show that

$$\left\| \nabla^m e^{t\Delta} e^{i\tau \frac{D_3}{|D|}} f \right\|_{L^4} \leq Ct^{-\frac{3}{2}(1-\frac{1}{4})-\frac{m}{2}} (1 + |\tau|)^{-\frac{3}{4}} (\log(e + |\tau|))^{\frac{1}{4}} \|f\|_{L^1},$$

and the logarithmic correction may be removed by using the weak Lebesgue space, that is,

$$\left\| \nabla^m e^{t\Delta} e^{i\tau \frac{D_3}{|D|}} f \right\|_{L^{4,\infty}} \leq Ct^{-\frac{3}{2}(1-\frac{1}{4})-\frac{m}{2}} (1 + |\tau|)^{-\frac{3}{4}} \|f\|_{L^1}.$$

Observe that our faster decay rates are shown to be sharp since it is possible to establish the lower bound estimates; see Theorem 1.4 below for the related topic. This is another novelty of the current work, since as far as we know there seems to be no previous results on the lower decay estimates for incompressible rotating Navier–Stokes equations.

Applying the above linear analysis to the nonlinear solutions with critical Sobolev regularity, we obtain the first main result of this paper.

Theorem 1.1. *Let $2 \leq p \leq \infty$ and $m \in \mathbb{N} \cup \{0\}$. Let $u_0 \in \dot{H}^{\frac{1}{2}}(\mathbb{R}^3) \cap L^1(\mathbb{R}^3)$ satisfy $\operatorname{div} u_0 = 0$. Then, there exists a positive constant $\Omega_0 = \Omega_0(u_0)$ such that for any $\Omega \in \mathbb{R}$ with $|\Omega| \geq \Omega_0$, (1.1) possesses a unique mild solution $u \in C([0, \infty); \dot{H}^{\frac{1}{2}}(\mathbb{R}^3)) \cap C((0, \infty); L^\infty(\mathbb{R}^3))$ and it holds*

$$\|\nabla^m u(t)\|_{L^p} = \begin{cases} o\left(t^{-\frac{3}{2}(1-\frac{1}{p})-\frac{m}{2}} (1 + |\Omega|t)^{-\frac{3}{2}(1-\frac{2}{p})}\right) & (2 \leq p < 4), \\ o\left(t^{-\frac{3}{2}(1-\frac{1}{4})-\frac{m}{2}} (1 + |\Omega|t)^{-\frac{3}{4}} (\log(e + |\Omega|t))^{\frac{1}{4}}\right) & (p = 4), \\ o\left(t^{-\frac{3}{2}(1-\frac{1}{p})-\frac{m}{2}} (1 + |\Omega|t)^{-(1-\frac{1}{p})}\right) & (4 < p \leq \infty) \end{cases} \quad (1.7)$$

as $t \rightarrow \infty$.

As for the classical Navier–Stokes equations, the small order decay rate may be improved to the additional $t^{-\frac{1}{2}}$ -faster decay estimate, provided that $|x|u_0(x)$ is integrable. In the following theorem, we state not only this but also removal of the logarithmic correction for $p = 4$.

Theorem 1.2. *Let $u_0 \in \dot{H}^{\frac{1}{2}}(\mathbb{R}^3) \cap L^1(\mathbb{R}^3)$ satisfy $\operatorname{div} u_0 = 0$ and additionally assume $|x|u_0(x) \in L^1(\mathbb{R}_x^3)$. Then, for every $2 \leq p \leq \infty$ and $m \in \mathbb{N} \cup \{0\}$, there exists a positive constant $C = C(m, p, u_0)$ such that the solution u to (1.1) with $|\Omega| \geq \Omega_0$, constructed in Theorem 1.1, satisfies the 1/2-enhanced decay estimates*

$$\|\nabla^m u(t)\|_{L^p} \leq \begin{cases} Ct^{-\frac{3}{2}(1-\frac{1}{p})-\frac{m+1}{2}} (1 + |\Omega|t)^{-\frac{3}{2}(1-\frac{2}{p})} & (2 \leq p < 4), \\ Ct^{-\frac{3}{2}(1-\frac{1}{4})-\frac{m+1}{2}} (1 + |\Omega|t)^{-\frac{3}{4}} (\log(e + |\Omega|t))^{\frac{1}{4}} & (p = 4), \\ Ct^{-\frac{3}{2}(1-\frac{1}{p})-\frac{m+1}{2}} (1 + |\Omega|t)^{-(1-\frac{1}{p})} & (4 < p \leq \infty) \end{cases} \quad (1.8)$$

for all $t \geq 1$. Furthermore, the logarithmic correction for $p = 4$ may be removed in $L^{4,\infty}$ framework:

$$\|\nabla^m u(t)\|_{L^{4,\infty}} \leq C t^{-\frac{3}{2}(1-\frac{1}{4})-\frac{m+1}{2}} (1 + |\Omega|t)^{-\frac{3}{4}}. \quad (1.9)$$

Remark 1.3. For the threshold case $p = 4$, the strength of logarithmic correction is actually controlled by the interpolation in Lorentz spaces:

$$\|u(t)\|_{L^{4,q}} \leq C t^{-\frac{3}{2}(1-\frac{1}{4})-\frac{m+1}{2}} (1 + |\Omega|t)^{-\frac{3}{4}} (\log(e + |\Omega|t))^{\frac{1}{q}}, \quad 1 \leq q \leq \infty,$$

which can be easily obtained by slightly modifying the linear analysis.

Next, we claim that the decay estimates (1.8) and (1.9) are optimal.

Theorem 1.4. Let $\Omega = 1$ and $2 \leq p \leq \infty$. Then, there exist $u_0 \in \mathcal{S}(\mathbb{R}^3)$ with $\operatorname{div} u_0 = 0$ and positive constants $c = c(p)$, $T = T(p)$ such that the associated mild solution u to (1.1) satisfies

$$\begin{aligned} \|u(t)\|_{L^p} &\geq \begin{cases} c t^{-\frac{3}{2}(1-\frac{1}{p})-\frac{1}{2}} (1+t)^{-\frac{3}{2}(1-\frac{2}{p})} & (2 \leq p < 4), \\ c t^{-\frac{3}{2}(1-\frac{1}{4})-\frac{1}{2}} (1+t)^{-\frac{3}{4}} (\log(e+t))^{\frac{1}{4}} & (p = 4), \\ c t^{-\frac{3}{2}(1-\frac{1}{p})-\frac{1}{2}} (1+t)^{-(1-\frac{1}{p})} & (4 < p \leq \infty), \end{cases} \\ \|u(t)\|_{L^{4,\infty}} &\geq c t^{-\frac{3}{2}(1-\frac{1}{4})-\frac{1}{2}} (1+t)^{-\frac{3}{4}} \end{aligned}$$

for all $t \geq T$.

Remark 1.5. In Theorem 1.4, the initial datum is chosen explicitly by

$$u_0(x) := \varepsilon e^{-\frac{|x|^2}{4}} (x_2, -x_1, 0),$$

where $\varepsilon > 0$ is chosen sufficiently small; see Theorem 3.7.

We briefly explain why the L^p decay estimate (1.5) derived by the interpolation does not capture the true behavior for $2 < p < \infty$. As mentioned above, the oscillatory decay rate is controlled by the degeneracy of the phase $\xi_3/|\xi|$ and it is measured by the Hessian

$$h(\xi) := \det \nabla_\xi^2 \left(\frac{\xi_3}{|\xi|} \right) = \frac{|\xi_h|^2 \xi_3}{|\xi|^9}.$$

Along this function, we decompose the oscillatory kernel into dyadic degeneracy layers

$$e^{i\tau \frac{D_3}{|D|}} \varphi = \sum_{k \leq k_0} I_k(\tau), \quad I_k(\tau, x) = \frac{1}{(2\pi)^3} \int_{|h(\xi)| \simeq 2^k} e^{ix \cdot \xi} e^{i\tau \frac{\xi_3}{|\xi|}} \widehat{\varphi}(\xi) d\xi.$$

By the estimate

$$\left| \left\{ \xi \in \operatorname{supp} \widehat{\varphi} ; |h(\xi)| \simeq 2^k \right\} \right| \simeq \left| \left\{ \xi \in \operatorname{supp} \widehat{\varphi} ; |\xi_h|^2 |\xi_3| \simeq 2^k \right\} \right| \simeq 2^k,$$

and the standard stationary phase estimate, it is naturally expected that the following estimate holds:

$$\|I_k(\tau)\|_{L^\infty} \leq C \min \left\{ 2^k, |\tau|^{-\frac{3}{2}} 2^{-\frac{k}{2}} \right\}, \quad \|I_k(\tau)\|_{L^2} \leq C 2^{k/2}.$$

Interpolating these estimates yields

$$\|I_k(\tau)\|_{L^p} \leq C \min \left\{ 2^{(1-\frac{1}{p})k}, |\tau|^{-\frac{3}{2}(1-\frac{2}{p})} 2^{(\frac{2}{p}-\frac{1}{2})k} \right\}, \quad 2 \leq p \leq \infty. \quad (1.10)$$

The two bounds balance at $2^k \simeq |\tau|^{-1}$. The global L^∞ estimate is therefore governed by this transition layer and has size $|\tau|^{-1}$. Interpolating only this worst L^∞ bound, which provides (1.3), with the conserved L^2 norm (1.4), we treat the oscillatory integral as if it were as degenerate as the transition layer throughout the whole annulus $\operatorname{supp} \widehat{\varphi}$, and this is precisely why the known decay rate $|\tau|^{-(1-\frac{2}{p})}$ in (1.5) is not sharp. For $2 \leq p < 4$, the exponent $2/p - 1/2$ in (1.10) is positive, so the sum over the less degenerate layers

is dominated by the nondegenerate part of the annulus and gives $|\tau|^{-\frac{3}{2}(1-\frac{2}{p})}$; the more degenerate layers are smaller because of their shrinking Fourier volume. For $4 < p \leq \infty$, the same sum is concentrated near the transition scale $2^k \simeq |\tau|^{-1}$ and gives $|\tau|^{-(1-\frac{1}{p})}$. Thus, the sharp L^p decay structure with $2 < p < \infty$ includes not only the worst pointwise degeneracy but also the small size of the frequency region where that degeneracy of the phase occurs, and the latter effect contributes the faster decay. *This type of phenomenon does not appear in the isotropic case (see for instance [16]); rather, it is caused by the anisotropy of the phase function in the oscillatory integral. To the best of our knowledge, the present paper is the first one to identify and capture such a phenomenon.*

The rest of this paper is arranged as follows. In Section 2, we collect some preliminary material, including some basic facts about Besov spaces and some useful lemmas. In Section 3, we proceed beyond the above heuristic argument and develop a unified framework combining an improved analysis of oscillatory integrals with distribution function estimates, which may even provide the optimal logarithmic correction in L^4 estimate and also give $L^{4,\infty}$ estimate. In Section 4, we apply those sharp linear estimates to the nonlinear problem and prove the main results.

2. PRELIMINARIES

In this section, we explain some notations and correct lemmas that will be used throughout this paper.

2.1. Littlewood–Paley theory and function spaces. In this subsection, we recall some function spaces which will be used in the sequel. Let $\mathcal{S}(\mathbb{R}^3)$ be the space of all Schwartz functions on $\mathbb{R}^3$ and $\mathcal{S}'(\mathbb{R}^3)$ be the set of all tempered distributions on $\mathbb{R}^3$. For any $f \in \mathcal{S}'(\mathbb{R}^3)$, we define the Fourier transform and inverse Fourier transform by

$$\mathcal{F}[f](\xi) = \widehat{f}(\xi) = \int_{\mathbb{R}^3} e^{-ix \cdot \xi} f(x) dx, \quad \mathcal{F}^{-1}[f](x) = (2\pi)^{-3} \int_{\mathbb{R}^3} e^{ix \cdot \xi} f(\xi) d\xi.$$

We recall the definition of the Littlewood–Paley decomposition and homogeneous Besov spaces. Let $\phi \in C_c^\infty([0, \infty); [0, 1])$ satisfy $\text{supp } \phi \subset [2^{-1}, 2]$ and $\sum_{j \in \mathbb{Z}} \phi(2^{-j}r) = 1$ for all $r > 0$. Let us define the dyadic localization operators $\{\Delta_j\}_{j \in \mathbb{Z}}$ by

$$\Delta_j f := \mathcal{F}^{-1} \left[\widehat{\phi}_j(\xi) \widehat{f}(\xi) \right], \quad \widehat{\phi}_j(\xi) := \phi(2^{-j}|\xi|).$$

For any $s \in \mathbb{R}, 1 \leq p, q \leq \infty$, we define the homogeneous Besov space $\dot{B}_{p,q}^s(\mathbb{R}^3)$ by

$$\begin{aligned} \dot{B}_{p,q}^s(\mathbb{R}^3) &:= \left\{ f \in \mathcal{S}'(\mathbb{R}^3) / \mathcal{P}(\mathbb{R}^3); \|f\|_{\dot{B}_{p,q}^s} < \infty \right\}, \\ \|f\|_{\dot{B}_{p,q}^s} &:= \left\| \left\{ 2^{sj} \|\Delta_j f\|_{L^p} \right\}_{j \in \mathbb{Z}} \right\|_{\ell^q(\mathbb{Z})}, \end{aligned}$$

where $\mathcal{P}(\mathbb{R}^3)$ denotes the set of all polynomials on $\mathbb{R}^3$. The following embeddings are well-known:

$$\dot{B}_{p,1}^0(\mathbb{R}^3) \hookrightarrow L^p(\mathbb{R}^3) \hookrightarrow \dot{B}_{p,\infty}^0(\mathbb{R}^3), \quad 1 \leq p \leq \infty,$$

and

$$\dot{B}_{p_1,q}^{\frac{3}{p_1}}(\mathbb{R}^3) \hookrightarrow \dot{B}_{p_2,q}^{\frac{3}{p_2}}(\mathbb{R}^3), \quad 1 \leq p_1 \leq p_2 \leq \infty, \quad 1 \leq q \leq \infty.$$

See [5] for the details. In the particular case of $p = q = 2$, we may identify $\dot{B}_{2,2}^s(\mathbb{R}^3)$ with the homogeneous Sobolev space $\dot{H}^s(\mathbb{R}^3)$ in the sense of norm equivalence.

We summarize some basic results on Besov spaces. We begin with an estimate of heat flow on Besov spaces.

Lemma 2.1 ([24, Lemma 2.2]). *Let $1 \leq p \leq \infty$ and $s_1, s_2 \in \mathbb{R}$ with $s_1 < s_2$. Then, there exists a positive constant $C = C(s_1, s_2)$ such that*

$$\|e^{t\Delta}f\|_{\dot{B}_{p,1}^{s_2}} \leq Ct^{-\frac{s_2-s_1}{2}} \|f\|_{\dot{B}_{p,\infty}^{s_1}}$$

for all $t > 0$ and $f \in \dot{B}_{p,\infty}^{s_1}(\mathbb{R}^3)$.

Next, we recall a real interpolation inequality for Besov spaces.

Lemma 2.2 ([5, Proposition 2.22]). *Let $1 \leq p \leq \infty$ and $s_0, s_1 \in \mathbb{R}$ with $s_0 < s_1$. Let $0 < \theta < 1$. Then there exists a positive constant $C = C(s_1, s_2, \theta)$ such that*

$$\|f\|_{\dot{B}_{p,1}^{\theta s_1 + (1-\theta)s_0}} \leq C \|f\|_{\dot{B}_{p,\infty}^{s_1}}^\theta \|f\|_{\dot{B}_{p,\infty}^{s_0}}^{1-\theta}$$

for all $f \in \dot{B}_{p,\infty}^{s_1}(\mathbb{R}^3) \cap \dot{B}_{p,\infty}^{s_2}(\mathbb{R}^3)$.

Let us review the following basic para-product estimate in Besov spaces.

Lemma 2.3 ([6, Lemma 2.2]). *For $s > 0$, $1 \leq p, p_1, p_2, q \leq \infty$ with $1/p = 1/p_1 + 1/p_2$, there exists a positive constant $C = C(s, p, p_1, p_2, q)$ such that*

$$\|fg\|_{\dot{B}_{p,q}^s} \leq C \left(\|f\|_{L^{p_1}} \|g\|_{\dot{B}_{p_2,q}^s} + \|f\|_{\dot{B}_{p_2,q}^s} \|g\|_{L^{p_1}} \right)$$

for all $f, g \in L^{p_1}(\mathbb{R}^3) \cap \dot{B}_{p_2,q}^s(\mathbb{R}^3)$.

We remark that the proof of Lemma 2.3 can also be obtained from [5, Theorem 2.47, 2.52]. The details are omitted.

Finally, we recall the weak Lebesgue spaces. For $1 \leq p < \infty$, the weak Lebesgue space $L^{p,\infty}(\mathbb{R}^3)$ consists of all measurable functions f on $\mathbb{R}^3$ such that

$$\|f\|_{L^{p,\infty}} := \sup_{\lambda > 0} \lambda \left| \{x \in \mathbb{R}^3; |f(x)| > \lambda\} \right|^{\frac{1}{p}} < \infty,$$

It is well-known that $L^p(\mathbb{R}^3) \subsetneq L^{p,\infty}(\mathbb{R}^3)$ when $1 \leq p < \infty$. For more details on the function spaces mentioned above, we refer to [5, 15].

2.2. Stationary phase lemmas for oscillatory integrals. Let us prepare the estimates for oscillatory integrals.

Lemma 2.4. *Let $K \subset \mathbb{R}^3$ be a non-empty compact set. Let $\{\psi_\alpha\}_{\alpha \in A} \subset C^6(K; \mathbb{R})$ and $\{b_\alpha\}_{\alpha \in A} \subset C_c^4(\mathbb{R}^3; \mathbb{R})$ be families of phase and amplitude functions satisfying*

$$\inf_{\alpha \in A} \inf_{\xi \in K} |\det \nabla^2 \psi_\alpha(\xi)| > 0, \quad \bigcup_{\alpha \in A} \text{supp } b_\alpha \subset K,$$

and

$$\sum_{m=0}^6 \sup_{\alpha \in A} \|\nabla^m \psi_\alpha\|_{L^\infty(K)} < \infty, \quad \sum_{m=0}^4 \sup_{\alpha \in A} \|\nabla^m b_\alpha\|_{L^\infty(K)} < \infty.$$

Then, there exists a positive constant C such that

$$\sup_{\alpha \in A} \sup_{x \in \mathbb{R}^3} \left| \int_{\mathbb{R}^3} e^{ix \cdot \xi} e^{i\tau \psi_\alpha(\xi)} b_\alpha(\xi) d\xi \right| \leq C |\tau|^{-\frac{3}{2}}$$

for all $\tau \in \mathbb{R}$ with $|\tau| \geq 1$.

One may prove Lemma 2.4 along the same argument as the standard stationary phase estimate; see [18, Theorem 7.7.1] or [28, Proposition 2.5, page 329] for instance. Next, we review the asymptotic expansion of oscillatory integrals.

Lemma 2.5 ([18, Theorem 7.7.6]). *Let $K \subset \mathbb{R}^3$ be a non-empty compact set. Let $\psi = \psi(y, \xi) \in C^\infty(K \times \mathbb{R}^3; \mathbb{R})$ and $a = a(y, \xi) \in C_c^\infty(K \times \mathbb{R}^3)$. Assume that there exists a $\Xi \in C^\infty(K; \mathbb{R}^3)$ such that*

- For $y \in K$, $\xi = \Xi(y)$ is a unique stationary point for $\mathbb{R}^3 \ni \xi \mapsto \psi(y, \xi) \in \mathbb{R}$ on $\text{supp } a(y, \cdot)$.
- It holds $\det \nabla_\xi^2 \psi(y, \Xi(y)) \neq 0$ for all $y \in K$.

Then, there exists a positive constant $C = C(K) > 0$ such that

$$\left| \frac{1}{(2\pi)^3} \int_{\mathbb{R}^3} e^{i\tau\psi(y, \xi)} a(y, \xi) d\xi - \frac{e^{i\tau\psi(y, \Xi(y))} e^{\frac{\pi}{4} i \text{sgn } \nabla_\xi^2 \psi(y, \Xi(y))}}{(2\pi\tau)^{\frac{3}{2}} |\det \nabla_\xi^2 \psi(y, \Xi(y))|^{\frac{1}{2}}} a(y, \Xi(y)) \right| \leq C\tau^{-\frac{5}{2}}$$

for all $y \in K$ and $\tau \geq 1$. Here, the signature of a real symmetric matrix is defined as the number of its positive eigenvalues minus the number of its negative eigenvalues.

3. LINEAR ANALYSIS

The aim of this section is to derive the sharp decay estimates for the Stokes–Coriolis semigroup:

$$\mathcal{T}_\Omega(t)u_0 = e^{t(\Delta - \Omega \mathbb{P}_{e_3} \times \mathbb{P})} u_0 = e^{t\Delta} e^{it\Omega \frac{D_3}{|D|}} P_+(D)u_0 + e^{t\Delta} e^{-it\Omega \frac{D_3}{|D|}} P_-(D)u_0.$$

where the Fourier multiplier $P_\pm(D) = \mathcal{F}^{-1} P_\pm(\xi) \mathcal{F}$ is defined in (1.2).

3.1. Refined decay estimates. To simplify the presentation, we introduce the following notation.

Definition 3.1. For $2 \leq p \leq \infty$ and $\tau \in \mathbb{R}$, we define

$$\mathcal{D}_p(\tau) := \begin{cases} (1 + |\tau|)^{-\frac{3}{2}(1 - \frac{2}{p})} & \text{for } 2 \leq p < 4, \\ (1 + |\tau|)^{-\frac{3}{4}} (\log(e + |\tau|))^{\frac{1}{4}} & \text{for } p = 4, \\ (1 + |\tau|)^{-(1 - \frac{1}{p})} & \text{for } 4 < p \leq \infty. \end{cases}$$

The goal of this subsection is to establish the following theorem.

Theorem 3.2. For $2 \leq p \leq \infty$, $m \in \mathbb{N} \cup \{0\}$, the following statements hold true.

- (1) For any $u_0 \in L^1(\mathbb{R}^3)$ with $\text{div } u_0 = 0$, it holds

$$\|\nabla^m \mathcal{T}_\Omega(t)u_0\|_{L^p} = o\left(t^{-\frac{3}{2}(1 - \frac{1}{p}) - \frac{m}{2}} \mathcal{D}_p(|\Omega|t)\right)$$

as $t \rightarrow \infty$.

- (2) If $|x|u_0(x) \in L^1(\mathbb{R}^3)$ with $\text{div } u_0 = 0$, then there exists a positive constant $C = C(p, m)$ such that

$$\|\nabla^m \mathcal{T}_\Omega(t)u_0\|_{L^p} \leq Ct^{-\frac{3}{2}(1 - \frac{1}{p}) - \frac{m+1}{2}} \mathcal{D}_p(|\Omega|t) \| |x|u_0(x) \|_{L^1(\mathbb{R}_x^3)}$$

for all $t > 0$.

We first focus on the dispersive effect of the evolution group $\{e^{i\tau \frac{D_3}{|D|}}\}_{\tau \in \mathbb{R}}$ in L^p and show the following proposition.

Proposition 3.3. There exists a positive constant C such that

$$\begin{aligned} \left\| \Delta_j e^{i\tau \frac{D_3}{|D|}} f \right\|_{L^p} &\leq C 2^{3(1 - \frac{1}{p})j} \mathcal{D}_p(\tau) \|\Delta_j f\|_{L^1}, \\ \left\| \Delta_j e^{i\tau \frac{D_3}{|D|}} f \right\|_{L^{4,\infty}} &\leq C 2^{3(1 - \frac{1}{4})j} (1 + |\tau|)^{-\frac{3}{4}} \|\Delta_j f\|_{L^1} \end{aligned}$$

for all $\tau \in \mathbb{R}$, $j \in \mathbb{Z}$ and $f \in \mathcal{S}'(\mathbb{R}^3)$ with $\Delta_j f \in L^1(\mathbb{R}^3)$.

Let us focus on the estimates for the oscillatory integral:

$$e^{i\tau \frac{D_3}{|D|}} \varphi(x) = \frac{1}{(2\pi)^3} \int_{\mathbb{R}^3} e^{ix \cdot \xi} e^{i\tau \Psi(\xi)} \widehat{\varphi}(\xi) d\xi,$$

where we have defined

$$\widehat{\varphi}(\xi) := \widehat{\varphi}_{-1}(\xi) + \widehat{\varphi}_0(\xi) + \widehat{\varphi}_1(\xi), \quad \Psi(\xi) := \frac{\xi_3}{|\xi|}.$$

Note that $\widehat{\varphi}$ is supported in $\{\xi \in \mathbb{R}^3; 1/4 \leq |\xi| \leq 4\}$ and it holds

$$\det \nabla^2 \Psi(\xi) = \frac{|\xi_h|^2 \xi_3}{|\xi|^9}.$$

Let $\varrho \in C_c^\infty(\mathbb{R}; [0, 1])$ be an even function satisfying $\varrho(s) = 1$ for $|s| \leq 1/8$ and $\varrho(s) = 0$ for $|s| \geq 1/4$. We define

$$\rho^h(\xi) := \varrho\left(\frac{|\xi_h|}{|\xi|}\right), \quad \rho^v(\xi) := \varrho\left(\frac{\xi_3}{|\xi|}\right), \quad \rho^0(\xi) := 1 - \rho^h(\xi) - \rho^v(\xi)$$

for $\xi \in \mathbb{R}^3 \setminus \{0\}$. Observe that the supports of ρ^h and ρ^v are disjoint, and thus it holds $0 \leq \rho_0(\xi) \leq 1$. Let us define

$$\begin{aligned} I^0(\tau, x) &:= \frac{1}{(2\pi)^3} \int_{\mathbb{R}^3} e^{ix \cdot \xi} e^{i\tau \Psi(\xi)} a^0(\xi) d\xi, & a^0(\xi) &:= \widehat{\varphi}(\xi) \rho^0(\xi), \\ I_k^h(\tau, x) &:= \frac{1}{(2\pi)^3} \int_{\mathbb{R}^3} e^{ix \cdot \xi} e^{i\tau \Psi(\xi)} a_k^h(\xi) d\xi, & a_k^h(\xi) &:= \widehat{\varphi}(\xi) \rho^h(\xi) \phi(2^{-k} |\xi_h|^2), \\ I_k^v(\tau, x) &:= \frac{1}{(2\pi)^3} \int_{\mathbb{R}^3} e^{ix \cdot \xi} e^{i\tau \Psi(\xi)} a_k^v(\xi) d\xi, & a_k^v(\xi) &:= \widehat{\varphi}(\xi) \rho^v(\xi) \phi(2^{-k} \xi_3). \end{aligned}$$

We then see that

$$e^{i\tau \frac{D_3}{|D|}} \varphi(x) = I^0(\tau, x) + \sum_{k \leq 1} I_k^h(\tau, x) + \sum_{k \leq 1} I_k^v(\tau, x).$$

Lemma 3.4. *There exists a positive constant C such that*

$$\begin{aligned} \|I^0(\tau, \cdot)\|_{L^p} &\leq C |\tau|^{-\frac{3}{2}(1-\frac{2}{p})}, \\ \|I_k^\#(\tau, \cdot)\|_{L^p} &\leq C \min \left\{ 2^{(1-\frac{1}{p})k}, 2^{(\frac{2}{p}-\frac{1}{2})k} |\tau|^{-\frac{3}{2}(1-\frac{2}{p})} \right\} \end{aligned} \quad (3.1)$$

for all $\# \in \{h, v\}$, $k \leq 1$, $2 \leq p \leq \infty$, and $\tau \in \mathbb{R}$ with $|\tau| \geq 2$.

Proof. It follows from the Plancherel theorem that

$$\begin{aligned} \|I^0(\tau, \cdot)\|_{L^2} &= (2\pi)^{-\frac{3}{2}} \|a^0\|_{L^2}, \\ \|I_k^\#(\tau, \cdot)\|_{L^2} &= (2\pi)^{-\frac{3}{2}} \|a_k^\#\|_{L^2} \leq C 2^{\frac{1}{2}k}. \end{aligned}$$

Hence, by the interpolation, the proof is reduced to the case of $p = \infty$.

Step 1. L^∞ estimates for I^0 . Since it holds $|\det \nabla^2 \Psi(\xi)| = |\xi_h|^2 |\xi_3| / |\xi|^9 \geq 2^{-21}$ for $\xi \in \text{supp } a^0$, we see by Lemma 2.4 that

$$\|I^0(\tau, \cdot)\|_{L^\infty} \leq C |\tau|^{-\frac{3}{2}}.$$

Step 2. L^∞ estimates for I_k^h . We see that

$$\|I_k^h(\tau, \cdot)\|_{L^\infty} \leq \frac{1}{(2\pi)^3} \left| \text{supp } a_k^h \right| \leq C 2^k. \quad (3.2)$$

It follows from the change of variable $(\eta_h, \eta_3) = (2^{-\frac{1}{2}k} \xi_h, \xi_3)$ that

$$I_k^h(\tau, x) = \frac{2^k}{(2\pi)^3} \sum_{\sigma \in \{\pm\}} e^{\sigma i \tau} \int_{\mathbb{R}^3} e^{i(2^{\frac{1}{2}k} x_h, x_3) \cdot \eta} e^{-\sigma i 2^k \tau \Psi_{\sigma, k}^h(\eta)} b_{\sigma, k}^h(\eta) d\eta,$$

where we have defined

$$\begin{aligned}\Psi_{\pm,k}^h(\eta) &:= 2^{-k} \left(1 \mp \Psi \left(2^{\frac{1}{2}k} \eta_h, \eta_3 \right) \right) \\ &= |\eta_h|^2 \left(2^k |\eta_h|^2 + \eta_3^2 \right)^{-\frac{1}{2}} \left\{ |\eta_3| + \left(2^k |\eta_h|^2 + \eta_3^2 \right)^{\frac{1}{2}} \right\}^{-1}, \\ b_{\pm,k}^h(\eta) &:= \widehat{\varphi} \left(2^{\frac{1}{2}k} \eta_h, \eta_3 \right) \varrho \left(2^{\frac{1}{2}k} |\eta_h| \left(2^k |\eta_h|^2 + \eta_3^2 \right)^{-\frac{1}{2}} \right) \phi(|\eta_h|^2) \mathbf{1}_{(0,\infty)}(\pm \eta_3).\end{aligned}$$

Elementary computations yield

$$\begin{aligned}\bigcup_{k \leq 1} \text{supp } b_{\pm,k}^h &\subset K^h := \left\{ \eta \in \mathbb{R}^3 ; \frac{1}{2} \leq |\eta_h|^2 \leq 2, \frac{\sqrt{15}}{16} \leq |\eta_3| \leq 4 \right\}, \\ \sum_{m=0}^6 \sup_{k \leq 1} \left\| \nabla^m \Psi_{\pm,k}^h \right\|_{L^\infty(K^h)} &< \infty, \quad \sum_{m=0}^4 \sup_{k \leq 1} \left\| \nabla^m b_{\pm,k}^h \right\|_{L^\infty(K^h)} < \infty,\end{aligned}$$

and

$$\left| \det \nabla^2 \Psi_{\pm,k}^h(\eta) \right| = |\eta_h|^2 |\eta_3| \left(2^k |\eta_h|^2 + \eta_3^2 \right)^{-\frac{9}{2}} \geq \sqrt{15} \cdot 2^{-23}, \quad k \leq 1, \eta \in K^h.$$

From Lemma 2.4, we see that

$$\left\| I_k^h(\tau, \cdot) \right\|_{L^\infty} \leq C 2^k \left(2^k |\tau| \right)^{-\frac{3}{2}} = C 2^{-\frac{1}{2}k} |\tau|^{-\frac{3}{2}}. \quad (3.3)$$

Gathering (3.2) and (3.3), we obtain

$$\left\| I_k^h(\tau, \cdot) \right\|_{L^\infty} \leq C \min \left\{ 2^k, 2^{-\frac{1}{2}k} |\tau|^{-\frac{3}{2}} \right\}.$$

Step 3. L^∞ estimates for I_k^v . For the estimates of I_k^v , we have

$$\left\| I_k^v(\tau, \cdot) \right\|_{L^\infty} \leq \frac{1}{(2\pi)^3} |\text{supp } a_k^v| \leq C 2^k \quad (3.4)$$

It holds by the change of variable $(\eta_h, \eta_3) = (\xi_h, 2^{-k} \xi_3)$ that

$$I_k^v(\tau, x) = \frac{2^k}{(2\pi)^3} \int_{\mathbb{R}^3} e^{i(x_h, 2^k x_3) \cdot \eta} e^{i 2^k \tau \Psi_k^v(\eta)} b_k^v(\eta) d\eta,$$

where we have set

$$\begin{aligned}\Psi_k^v(\eta) &:= \eta_3 \left(|\eta_h|^2 + 2^{2k} \eta_3^2 \right)^{-\frac{1}{2}}, \\ b_k^v(\eta) &:= \widehat{\varphi}(\eta_h, 2^k \eta_3) \varrho \left(2^k \eta_3 \left(|\eta_h|^2 + 2^{2k} \eta_3^2 \right)^{-\frac{1}{2}} \right) \phi(\eta_3).\end{aligned}$$

By direct calculations, it holds

$$\begin{aligned}\bigcup_{k \leq 1} \text{supp } b_k^v &\subset K^v := \left\{ \eta \in \mathbb{R}^3 ; \frac{\sqrt{15}}{16} \leq |\eta_h| \leq 4, \frac{1}{2} \leq |\eta_3| \leq 2 \right\}, \\ \sum_{m=0}^6 \sup_{k \leq 1} \left\| \nabla^m \Psi_k^v \right\|_{L^\infty(K^v)} &< \infty, \quad \sum_{m=0}^4 \sup_{k \leq 1} \left\| \nabla^m b_k^v \right\|_{L^\infty(K^v)} < \infty,\end{aligned}$$

and

$$\left| \det \nabla^2 \Psi_k^v(\eta) \right| = |\eta_h|^2 |\eta_3| \left(|\eta_h|^2 + 2^{2k} \eta_3^2 \right)^{-\frac{9}{2}} \geq 15 \cdot 2^{-27}, \quad k \leq 1, \eta \in K^v.$$

Then, by Lemma 2.4, we have

$$\left\| I_k^v(\tau, \cdot) \right\|_{L^\infty} \leq C 2^k \left(2^k |\tau| \right)^{-\frac{3}{2}} = C 2^{-\frac{1}{2}k} |\tau|^{-\frac{3}{2}}. \quad (3.5)$$

Combining (3.4) and (3.5), we have

$$\|I_k^v(\tau, \cdot)\|_{L^\infty} \leq C \min \left\{ 2^k, 2^{-\frac{1}{2}k} |\tau|^{-\frac{3}{2}} \right\}$$

and we complete the proof. $\square$

Using Lemma 3.4, we obtain the following L^p decay estimates for $e^{i\tau \frac{D_3}{|D|}} \varphi$.

Lemma 3.5. *For $2 \leq p \leq \infty$, there exists a positive constant $C = C(p)$ such that*

$$\left\| e^{i\tau \frac{D_3}{|D|}} \varphi \right\|_{L^p} \leq C \mathcal{D}_p(\tau), \quad \left\| e^{i\tau \frac{D_3}{|D|}} \varphi \right\|_{L^{4,\infty}} \leq C(1 + |\tau|)^{-\frac{3}{4}}.$$

for all $\tau \in \mathbb{R}$.

Proof. Since we see for $2 \leq p \leq \infty$ that

$$\left\| e^{i\tau \frac{D_3}{|D|}} \varphi \right\|_{L^p} \leq \left\| e^{i\tau \frac{D_3}{|D|}} \varphi \right\|_{L^2}^{\frac{2}{p}} \left\| e^{i\tau \frac{D_3}{|D|}} \varphi \right\|_{L^\infty}^{1-\frac{2}{p}} \leq C \|\varphi\|_{L^2}^{\frac{2}{p}} \|\widehat{\varphi}\|_{L^1}^{1-\frac{2}{p}},$$

it suffices to consider only for sufficiently large $|\tau|$. Moreover, (3.1) implies that it is enough to focus on the estimates for

$$I^\#(\tau, x) := \sum_{k \leq 1} I_k^\#(\tau, x) = \frac{1}{(2\pi)^3} \int_{\mathbb{R}^3} e^{ix \cdot \xi} e^{i\tau \Psi(\xi)} a_k^\#(\xi) d\xi, \quad \# \in \{h, v\}.$$

Step 1. The L^p estimates with $p \neq 4$. It follows from Lemma 3.4 that

$$\begin{aligned} \left\| I^\#(\tau, \cdot) \right\|_{L^p} &\leq \sum_{k \leq 1} \left\| I_k^\#(\tau, \cdot) \right\|_{L^p} \\ &\leq C \sum_{k \leq 1} \min \left\{ 2^{(1-\frac{1}{p})k}, 2^{(\frac{2}{p}-\frac{1}{2})k} |\tau|^{-\frac{3}{2}(1-\frac{2}{p})} \right\} \\ &\leq C \sum_{2^k \leq |\tau|^{-1}} 2^{(1-\frac{1}{p})k} + C |\tau|^{-\frac{3}{2}(1-\frac{2}{p})} \sum_{|\tau|^{-1} < 2^k \leq 2} 2^{(\frac{2}{p}-\frac{1}{2})k} \\ &\leq C \mathcal{D}_p(\tau). \end{aligned}$$

Step 2. The $L^{4,\infty}$ estimates. Let δ be a positive constant to be determined later and define

$$\mathbb{Z}_{\leq 1} := \{k \in \mathbb{Z} ; k \leq 1\}, \quad E(\tau, \lambda) := \left\{ k \in \mathbb{Z}_{\leq 1} ; \min \left\{ 2^k, 2^{-\frac{1}{2}k} |\tau|^{-\frac{3}{2}} \right\} > \delta \lambda \right\}.$$

We then see that

$$\begin{aligned} \left\| \sum_{k \in \mathbb{Z}_{\leq 1} \setminus E(\tau, \lambda)} I_k^\#(\tau, \cdot) \right\|_{L^\infty} &\leq \sum_{k \in \mathbb{Z}_{\leq 1} \setminus E(\tau, \lambda)} \left\| I_k^\#(\tau, \cdot) \right\|_{L^\infty} \\ &\leq C \sum_{k \in \mathbb{Z}_{\leq 1} \setminus E(\tau, \lambda)} \min \left\{ 2^k, 2^{-\frac{1}{2}k} |\tau|^{-\frac{3}{2}} \right\} \\ &\leq C \delta \lambda. \end{aligned}$$

Since the above constant C may be chosen independent of δ , choosing $\delta = (2C)^{-1}$ yields

$$\left\| \sum_{k \in \mathbb{Z}_{k \leq 1} \setminus E(\tau, \lambda)} I_k^\#(\tau, \cdot) \right\|_{L^\infty} \leq \frac{\lambda}{2}.$$

This implies

$$\left\{ x \in \mathbb{R}^3 ; \left| I^\#(\tau, x) \right| > \lambda \right\} \subset \left\{ x \in \mathbb{R}^3 ; \left| \sum_{k \in E(\tau, \lambda)} I_k^\#(\tau, \cdot) \right| > \frac{\lambda}{2} \right\} \quad (3.6)$$

Thus, we see by (3.6), Chebyshev inequality, Lemma 3.4 with $p = 2$, and the almost orthogonality that

$$\begin{aligned}
 \left| \left\{ x \in \mathbb{R}^3 ; \left| I^\#(\tau, x) \right| > \lambda \right\} \right| &\leq \left| \left\{ x \in \mathbb{R}^3 ; \left| \sum_{k \in E(\tau, \lambda)} I_k^\#(\tau, \cdot) \right| > \frac{\lambda}{2} \right\} \right| \\
 &\leq \frac{4}{\lambda^2} \left\| \sum_{k \in E(\tau, \lambda)} I_k^\#(\tau, \cdot) \right\|_{L^2}^2 \\
 &\leq \frac{C}{\lambda^2} \sum_{k \in E(\tau, \lambda)} \left\| I_k^\#(\tau, \cdot) \right\|_{L^2}^2 \\
 &\leq \frac{C}{\lambda^2} \sum_{2^k < \delta^{-2} |\tau|^{-3} \lambda^{-2}} 2^k \\
 &\leq C |\tau|^{-3} \lambda^{-4}.
 \end{aligned} \tag{3.7}$$

Hence, we obtain

$$\left\| I^\#(\tau, \cdot) \right\|_{L^{4, \infty}} = \sup_{\lambda > 0} \lambda \left| \left\{ x \in \mathbb{R}^3 ; \left| I^\#(\tau, x) \right| > \lambda \right\} \right|^{\frac{1}{4}} \leq C |\tau|^{-\frac{3}{4}}.$$

Step 3. The L^4 estimate. The L^∞ estimate for $I^\#$ yields that

$$\left| \left\{ x \in \mathbb{R}^3 ; \left| I^\#(\tau, x) \right| > \lambda \right\} \right| = 0 \quad \text{for } \lambda > C |\tau|^{-1}. \tag{3.8}$$

By the Chebyshev inequality and the Plancherel theorem, we have

$$\begin{aligned}
 \left| \left\{ x \in \mathbb{R}^3 ; \left| I^\#(\tau, x) \right| > \lambda \right\} \right| &\leq \left\| I^\#(\tau, \cdot) \right\|_{L^2}^2 \lambda^{-2} \\
 &= (2\pi)^{-3} \left\| \widehat{\varphi} \rho^\# \right\|_{L^2}^2 \lambda^{-2} = C \lambda^{-2}.
 \end{aligned} \tag{3.9}$$

Hence, from (3.7), (3.8), (3.9), and the layer-cake theorem, it follows that

$$\begin{aligned}
 \left\| I^\#(\tau, \cdot) \right\|_{L^4}^4 &= 4 \int_0^{C |\tau|^{-1}} \lambda^4 \left| \left\{ x \in \mathbb{R}^3 ; \left| I^\#(\tau, x) \right| > \lambda \right\} \right| \frac{d\lambda}{\lambda} \\
 &\leq C \int_0^{|\tau|^{-\frac{3}{2}}} \lambda d\lambda + C |\tau|^{-3} \int_{|\tau|^{-\frac{3}{2}}}^{C |\tau|^{-1}} \frac{d\lambda}{\lambda} \\
 &\leq C |\tau|^{-3} \log |\tau|.
 \end{aligned}$$

This completes the proof. $\square$

Now, we are in a position to prove Proposition 3.3.

Proof of Proposition 3.3. By the almost orthogonality of the Littlewood–Paley decomposition, we have

$$\Delta_j e^{i\tau \frac{D_3}{|D|}} f(x) = 2^{3j} e^{i\tau \frac{D_3}{|D|}} \varphi(2^j \cdot) * \Delta_j f.$$

Thus, we see by the Young inequality and Lemma 3.5 that

$$\begin{aligned}
 \left\| \Delta_j e^{i\tau \frac{D_3}{|D|}} f \right\|_{L^p} &\leq 2^{3j} \left\| e^{i\tau \frac{D_3}{|D|}} \varphi(2^j \cdot) \right\|_{L^p} \left\| \Delta_j f \right\|_{L^1} \\
 &= 2^{3(1-\frac{1}{p})j} \left\| e^{i\tau \frac{D_3}{|D|}} \varphi \right\|_{L^p} \left\| \Delta_j f \right\|_{L^1} \\
 &\leq C 2^{3(1-\frac{1}{p})j} \mathcal{D}_p(\tau) \left\| \Delta_j f \right\|_{L^1}.
 \end{aligned}$$

For the $L^{4,\infty}$ estimate, we have by the same token that

$$\begin{aligned} \left\| \Delta_j e^{i\tau \frac{D_3}{|D|}} f \right\|_{L^{4,\infty}} &\leq C 2^{3j} \left\| e^{i\tau \frac{D_3}{|D|}} \varphi(2^j \cdot) \right\|_{L^{4,\infty}} \|\Delta_j f\|_{L^1} \\ &= C 2^{3(1-\frac{1}{4})j} \left\| e^{i\tau \frac{D_3}{|D|}} \varphi \right\|_{L^{4,\infty}} \|\Delta_j f\|_{L^1} \\ &\leq C 2^{3(1-\frac{1}{4})j} (1 + |\tau|)^{-\frac{3}{4}} \|\Delta_j f\|_{L^1}. \end{aligned}$$

Thus, we complete the proof. $\square$

Next, we provide the L^p decay estimates for the semigroup $\{e^{t\Delta} e^{i\tau\Psi(D)}\}_{t>0}$.

Proposition 3.6. *Let $2 \leq p \leq \infty$, $m \in \mathbb{N} \cup \{0\}$. Then, there exists a positive constant $C = C(p, m)$ such that*

$$\begin{aligned} \left\| \nabla^m e^{t\Delta} e^{i\tau\Psi(D)} f \right\|_{L^p} &\leq C t^{-\frac{3}{2}(1-\frac{1}{p})-\frac{m}{2}} \mathcal{D}_p(\tau) \|f\|_{\dot{B}_{1,\infty}^0}, \\ \left\| \nabla^m e^{t\Delta} e^{i\tau\Psi(D)} f \right\|_{L^{4,\infty}} &\leq C t^{-\frac{3}{2}(1-\frac{1}{4})-\frac{m}{2}} (1 + |\tau|)^{-\frac{3}{4}} \|f\|_{\dot{B}_{1,\infty}^0} \end{aligned}$$

for all $t > 0$, $\tau \in \mathbb{R}$ and $f \in \dot{B}_{1,\infty}^0(\mathbb{R}^3)$.

Proof. By the Littlewood–Paley decomposition and the Bernstein inequality, we deduce from Proposition 3.3 that

$$\begin{aligned} \left\| \nabla^m e^{t\Delta} e^{i\tau\Psi(D)} f \right\|_{L^p} &\leq \sum_{j \in \mathbb{Z}} \left\| \nabla^m e^{t\Delta} \Delta_j e^{i\tau\Psi(D)} f \right\|_{L^p} \\ &\leq C \sum_{j \in \mathbb{Z}} 2^{mj} e^{-c2^{2j}t} \left\| \Delta_j e^{i\tau\Psi(D)} f \right\|_{L^p} \\ &\leq C \mathcal{D}_p(\tau) \sum_{j \in \mathbb{Z}} 2^{\{m+3(1-\frac{1}{p})\}j} e^{-c2^{2j}t} \|\Delta_j f\|_{L^1} \\ &\leq C t^{-\frac{3}{2}(1-\frac{1}{p})-\frac{m}{2}} \mathcal{D}_p(\tau) \|f\|_{\dot{B}_{1,\infty}^0}. \end{aligned}$$

The $L^{4,\infty}$ estimate is obtained similarly. Thus, we complete the proof. $\square$

Proof of Theorem 3.2. Let

$$K_\Omega^\pm(t, x) := \mathcal{F}^{-1} \left[e^{\pm i\Omega t \frac{\xi_3}{|\xi|}} e^{-t|\xi|^2} P_\pm(\xi) \right] (x).$$

Since $u_0 \in L^1(\mathbb{R}^3)$ and $\operatorname{div} u_0 = 0$, the fact shown by Miyakawa [25] yields

$$\int_{\mathbb{R}^3} u_0(y) dy = 0.$$

Thus, we have

$$\begin{aligned} \mathcal{T}_\Omega(t) u_0(x) &= \sum_{\sigma \in \{\pm\}} \int_{\mathbb{R}^3} K_\Omega^\sigma(t, x-y) u_0(y) dy \\ &= \sum_{\sigma \in \{\pm\}} \int_{\mathbb{R}^3} (K_\Omega^\sigma(t, x-y) - K_\Omega^\sigma(t, x)) u_0(y) dy, \end{aligned}$$

which implies

$$\|\nabla^m \mathcal{T}_\Omega(t) u_0\|_{L^p} \leq \sum_{\sigma \in \{\pm\}} \int_{\mathbb{R}^3} \|\nabla^m K_\Omega^\sigma(t, \cdot - y) - \nabla^m K_\Omega^\sigma(t, \cdot)\|_{L^p} |u_0(y)| dy. \quad (3.10)$$

For $R > 0$, we deduce from Proposition 3.6 that

$$\begin{aligned} \|\nabla^m \mathcal{T}_\Omega u_0\|_{L^p} &\leq \sum_{\sigma \in \{\pm\}} \int_{|y| \leq R} \int_0^1 \|\nabla^{m+1} K_\Omega^\sigma(t, \cdot - \theta y)\|_{L^p} d\theta |y| |u_0(y)| dy \\ &\quad + 2 \sum_{\sigma \in \{\pm\}} \|\nabla^m K_\Omega^\sigma(t, \cdot)\|_{L^p} \int_{|y| \geq R} |u_0(y)| dy \\ &\leq C R t^{-\frac{3}{2}(1-\frac{1}{p})-\frac{m+1}{2}} \mathcal{D}_p(|\Omega|t) \int_{\mathbb{R}^3} |u_0(y)| dy \\ &\quad + C t^{-\frac{3}{2}(1-\frac{1}{p})-\frac{m}{2}} \mathcal{D}_p(|\Omega|t) \int_{|y| \geq R} |u_0(y)| dy, \end{aligned}$$

which implies

$$\limsup_{t \rightarrow \infty} t^{\frac{3}{2}(1-\frac{1}{p})+\frac{m}{2}} \mathcal{D}_p(|\Omega|t)^{-1} \|\nabla^m \mathcal{T}_\Omega(t) u_0\|_{L^p} \leq C \int_{|y| \geq R} |u_0(y)| dy.$$

Letting $R \rightarrow \infty$, we complete the first assertion. For the second assertion, we have by (3.10) that

$$\begin{aligned} \|\nabla^m \mathcal{T}_\Omega u_0\|_{L^p} &\leq \sum_{\sigma \in \{\pm\}} \int_{\mathbb{R}^3} \int_0^1 \|\nabla^{m+1} K_\Omega^\sigma(t, \cdot - \theta y)\|_{L^p} d\theta |y| |u_0(y)| dy \\ &\leq C t^{-\frac{3}{2}(1-\frac{1}{p})-\frac{m+1}{2}} \mathcal{D}_p(|\Omega|t) \int_{\mathbb{R}^3} |y| |u_0(y)| dy. \end{aligned}$$

This completes the proof. $\square$

3.2. Sharpness of decay rates. In this subsection, we show that the L^p decay rates of the linear solution obtained in the above section is optimal.

Theorem 3.7. *Let $\Omega = 1$ and define a divergence-free vector field $u_0 \in \mathcal{S}(\mathbb{R}^3)$ by*

$$u_0(x) := -16^{-1} \pi^{-\frac{3}{2}} e^{-\frac{|x|^2}{4}} x^\perp, \quad x^\perp := (x_2, -x_1, 0)$$

for $x \in \mathbb{R}^3$. Then, for any $2 \leq p \leq \infty$, there exist positive constants $T_0 = T_0(p)$ and $c = c(p)$ such that

$$\begin{aligned} \|\mathcal{T}_1(t) u_0\|_{L^p} &\geq c t^{-\frac{3}{2}(1-\frac{1}{p})-\frac{1}{2}} \mathcal{D}_p(t), \\ \|\mathcal{T}_1(t) u_0\|_{L^{4,\infty}} &\geq c t^{-\frac{3}{2}(1-\frac{1}{4})-\frac{1}{2}} (1+t)^{-\frac{3}{4}} \end{aligned}$$

for all $t \geq T_0$.

Proof. Since a direct computation yields $\widehat{u_0}(\xi) = e^{-|\xi|^2} i\xi^\perp$, it holds

$$\begin{aligned} \mathcal{T}_1(t) u_0(x) &= \frac{1}{(2\pi)^3} \sum_{\sigma \in \{\pm\}} \int_{\mathbb{R}^3} e^{ix \cdot \xi} e^{\sigma i t \Psi(\xi)} e^{-(t+1)|\xi|^2} P_\sigma(\xi) i\xi^\perp d\xi \\ &= \frac{1}{(t+1)^2} U\left(t, t^{-1}(t+1)^{-\frac{1}{2}} x\right), \end{aligned}$$

where we have set

$$\begin{aligned} U(t, z) &:= \sum_{\sigma \in \{\pm\}} \frac{1}{(2\pi)^3} \int_{\mathbb{R}^3} e^{it\Phi^\sigma(y, \xi)} e^{-|\xi|^2} P_\sigma(\xi) i\xi^\perp d\xi, \\ \Phi^\pm(y, \xi) &:= y \cdot \xi \pm \Psi(\xi). \end{aligned}$$

Then, we have

$$\|\mathcal{T}_1(t) u_0\|_{L^p} = (t+1)^{-\frac{3}{2}(1-\frac{1}{p})-\frac{1}{2}} t^{\frac{3}{p}} \|U(t, \cdot)\|_{L^p},$$

which implies that it suffices to show

$$\|U(t, \cdot)\|_{L^p} \geq \begin{cases} ct^{-\frac{3}{2}} & (2 \leq p < 4), \\ ct^{-\frac{3}{2}}(\log t)^{\frac{1}{4}} & (p = 4), \\ ct^{-1-\frac{2}{p}} & (4 < p \leq \infty), \end{cases} \quad \|U\|_{L^{4,\infty}} \geq ct^{-\frac{3}{2}}.$$

Step 1. L^p estimates with $2 \leq p < 4$ and $L^{4,\infty}$ estimate. Let

$$K := \left\{ y \in \mathbb{R}^3 ; \frac{1}{2} \leq |y_h| \leq 1, \quad -1 \leq y_3 \leq -\frac{1}{2} \right\}.$$

Fix a $y \in K$. Then, the stationary points of $\Phi^\pm(y, \cdot)$ are given by

$$\{\xi \in \mathbb{R}^3 ; \nabla_\xi \Phi^\sigma(y, \xi) = 0\} = \begin{cases} \{\Xi^+(y), \Xi^-(y)\} & (\sigma = +), \\ \emptyset & (\sigma = -), \end{cases}$$

where we have put

$$\Xi^\pm(y) := \left(\pm \frac{y_1 y_3^2}{|y_h||y|^3}, \pm \frac{y_2 y_3^2}{|y_h||y|^3}, \mp \frac{|y_h| y_3}{|y|^3} \right).$$

Let $\chi \in C_c^\infty(\mathbb{R}^3; [0, 1])$ and $\chi(0) = 1$ and $\text{supp } \chi \subset \{\xi \in \mathbb{R}^3 ; |\xi| \leq 1\}$. We define

$$\chi^\pm(\xi, y) := \chi(1000(\xi - \Xi^\pm(y))), \quad \chi^r(\xi, y) := 1 - \chi^+(\xi, y) - \chi^-(\xi, y).$$

Along this decomposition, we decompose U^+ as

$$\begin{aligned} U(t, y) &= U^+(t, y) + U^-(t, y) + U^r(t, y), \\ U^\pm(t, y) &:= \frac{1}{(2\pi)^3} \int_{\mathbb{R}^3} e^{it\Phi^\pm(y, \xi)} e^{-|\xi|^2} P_\pm(\xi) i\xi^\perp \chi^\pm(\xi, y) d\xi, \\ U^r(t, y) &:= \frac{1}{(2\pi)^3} \int_{\mathbb{R}^3} e^{it\Phi^+(y, \xi)} e^{-|\xi|^2} P_+(\xi) i\xi^\perp \chi^r(\xi, y) d\xi \\ &\quad + \frac{1}{(2\pi)^3} \int_{\mathbb{R}^3} e^{it\Phi^-(y, \xi)} e^{-|\xi|^2} P_-(\xi) i\xi^\perp d\xi. \end{aligned}$$

Lemmas 2.5 yields

$$\|U^\pm(t, \cdot) - V^\pm(t, \cdot)\|_{L^\infty(K)} \leq Ct^{-\frac{5}{2}}.$$

for all $\sigma \in \{\pm\}$ and $t \geq 1$, where

$$V^\pm(t, y) := \frac{e^{it\Phi^\pm(y, \Xi^\pm(y))} e^{i\frac{\pi}{4} \text{sgn } \nabla^2 \Psi(\Xi^\pm(y))}}{(2\pi t)^{\frac{3}{2}} |\det \nabla^2 \Psi(\Xi^\pm(y))|^{\frac{1}{2}}} e^{-|\Xi^\pm(y)|^2} P_\pm(\Xi^\pm(y)) i\Xi^\pm(y)^\perp.$$

Since direct calculations show that

$$|\det \nabla^2 \Psi(\Xi^\pm(y))| = \frac{|y_h||y|^9}{|y_3|^4}, \quad \text{sgn } \nabla^2 \Psi(\Xi^\pm(y)) = \mp 1$$

and

$$|\Xi^\pm(y)|^2 = \frac{y_3^2}{|y|^4}, \quad P_\pm(\Xi^\pm(y)) i\Xi^\pm(y)^\perp = \frac{y_3^2}{2|y|^3} \left(\pm i \frac{y^\perp}{|y_h|} - \frac{y}{|y|} \right),$$

we may write $V(t, y) = V^+(t, y) + V^-(t, y)$ explicitly as

$$V(t, y) = \frac{1}{(2\pi t)^{\frac{3}{2}} |y_h|^{\frac{1}{2}} |y|^{\frac{15}{2}}} \exp\left(-\frac{y_3^2}{|y|^4}\right) \left(\sin \Theta(t, y) \frac{y^\perp}{|y_h|} + \cos \Theta(t, y) \frac{y}{|y|} \right),$$

where

$$\Theta(t, y) := \frac{|y_h|}{|y|} t - \frac{\pi}{4}.$$

Then, we see that

$$|V(t, y)| = \frac{1}{(2\pi t)^{\frac{3}{2}}} \frac{y_3^4}{|y_h|^{\frac{1}{2}} |y|^{\frac{15}{2}}} \exp\left(-\frac{y_3^2}{|y|^4}\right),$$

which provides

$$\|V(t, \cdot)\|_{\mathcal{L}^p(K)} = c_p t^{-\frac{3}{2}},$$

where $\mathcal{L}^p$ stands for L^p for $2 \leq p < 4$ and $L^{4,\infty}$ for $p = 4$. For the remainder term, since the phase function $\Phi^+(y, \cdot)$ has no stationary points on the support of $\chi^r(\cdot, y)$ and $\Phi^-(y, \cdot)$ does not possess stationary points for $y \in K$, performing integration by parts three times yields

$$\|U^r(t, \cdot)\|_{L^\infty(K)} \leq C t^{-3}.$$

Hence, we obtain

$$\begin{aligned} \|U(t, \cdot)\|_{\mathcal{L}^p} &\geq \|U(t, \cdot)\|_{\mathcal{L}^p(K)} \\ &\geq \|V(t, \cdot)\|_{\mathcal{L}^p(K)} - \sum_{\sigma \in \{\pm\}} \|U^\sigma(t, \cdot) - V^\sigma(t, \cdot)\|_{L^\infty(K)} |K|^{\frac{1}{p}} \\ &\quad - \|U^r(t, \cdot)\|_{L^\infty(K)} |K|^{\frac{1}{p}} \\ &\geq c_p t^{-\frac{3}{2}} - C t^{-\frac{5}{2}} - C t^{-3} \\ &\geq \frac{c_p}{2} t^{-\frac{3}{2}} \end{aligned}$$

for sufficiently large t .

Step 2. The L^p estimates with $4 < p \leq \infty$. Let us consider the vertical component

$$U_3(t, y) = \frac{1}{(2\pi)^3} \sum_{\sigma \in \{\pm\}} \frac{\sigma}{2} \int_{\mathbb{R}^3} e^{it\Phi^\sigma(y, \xi)} e^{-|\xi|^2} \frac{|\xi_h|^2}{|\xi|} d\xi$$

for $t \geq 1$ and $y \in \mathbb{R}^3$ with $1 \leq y_3 \leq 2$. Using the 3D polar coordinate $\xi = (r \sin \eta \cos \theta, r \sin \eta \sin \theta, r \cos \eta)$ and suitable rotation of the horizontal coordinate, we see that

$$\begin{aligned} U_3(t, y) &= \frac{1}{(2\pi)^3} \sum_{\sigma \in \{\pm\}} \frac{\sigma}{2} \int_0^\infty \int_0^\pi \int_{-\pi}^\pi e^{it\phi^\sigma(y, r, \eta, \theta)} e^{-r^2} r^3 \sin^3 \eta dr d\eta d\theta, \\ &= \frac{1}{(2\pi)^2} \sum_{\sigma \in \{\pm\}} \frac{\sigma}{2} \int_0^\infty r^3 e^{-r^2} \int_0^\pi J(tr|y_h| \sin \eta) e^{it(r y_3 \sigma 1) \cos \eta} \sin^3 \eta dr d\eta \\ &= \frac{1}{2(2\pi)^2} (I^+(t, y) - I^-(t, y)), \end{aligned}$$

where we have put

$$\begin{aligned} \phi^\pm(y, r, \eta, \theta) &:= r|y_h| \sin \eta \cos \theta + (r y_3 \pm 1) \cos \eta, \\ J(\rho) &:= \frac{1}{2\pi} \int_{-\pi}^\pi e^{i\rho \cos \theta} d\theta \end{aligned}$$

and

$$\begin{aligned} I^\pm(t, y) &:= \int_0^\infty r^3 e^{-r^2} H(tr|y_h|, t(r y_3 \pm 1)) dr, \\ H(\rho, \tau) &:= \int_0^\pi J(\rho \sin \eta) e^{i\tau \cos \eta} \sin^3 \eta d\eta \\ &= \int_{-1}^1 h(\rho, \zeta) e^{i\tau \zeta} d\zeta, \end{aligned}$$

$$h(\rho, \zeta) := (1 - \zeta^2)J\left(\rho\sqrt{1 - \zeta^2}\right).$$

Let us investigate the properties of H . By the integration by parts, we have

$$\begin{aligned} H(\rho, \tau) &= -\frac{1}{i\tau} \int_{-1}^1 \partial_\zeta h(\rho, \zeta) e^{i\tau\zeta} d\zeta \\ &= \frac{1}{\tau^2} \left[\partial_\zeta h(\rho, \zeta) e^{i\tau\zeta} \right]_{\zeta=-1}^{\zeta=1} - \frac{1}{\tau^2} \int_{-1}^1 \partial_\zeta^2 h(\rho, \zeta) e^{i\tau\zeta} d\zeta \\ &= \frac{4\cos\tau}{\tau^2} + \frac{2}{\tau^2} \int_{-1}^1 J(\rho\sqrt{1 - \zeta^2}) e^{i\tau\zeta} d\zeta \\ &\quad - \frac{\rho}{\tau^2} \int_{-1}^1 \frac{4\zeta^2 - 1}{\sqrt{1 - \zeta^2}} J'(\rho\sqrt{1 - \zeta^2}) e^{i\tau\zeta} d\zeta \\ &\quad - \frac{\rho^2}{\tau^2} \int_{-1}^1 \zeta^2 J''(\rho\sqrt{1 - \zeta^2}) e^{i\tau\zeta} d\zeta. \end{aligned}$$

From this, we see that

$$\begin{aligned} \partial_\rho H(\rho, \tau) &= -\frac{2\rho}{\tau^2} \int_{-1}^1 \frac{\zeta}{\sqrt{1 - \zeta^2}} J'(\rho\sqrt{1 - \zeta^2}) e^{i\tau\zeta} d\zeta \\ &\quad - \frac{1}{\tau^2} \int_{-1}^1 \frac{4\zeta^2 - 1}{\sqrt{1 - \zeta^2}} J'(\rho\sqrt{1 - \zeta^2}) e^{i\tau\zeta} d\zeta - \frac{\rho^2}{\tau^2} \int_{-1}^1 (4\zeta^2 - 1) J'(\rho\sqrt{1 - \zeta^2}) e^{i\tau\zeta} d\zeta \\ &\quad - \frac{2\rho}{\tau^2} \int_{-1}^1 \zeta^2 J''(\rho\sqrt{1 - \zeta^2}) e^{i\tau\zeta} d\zeta - \frac{\rho^2}{\tau^2} \int_{-1}^1 \zeta^2 \sqrt{1 - \zeta^2} J'''(\rho\sqrt{1 - \zeta^2}) e^{i\tau\zeta} d\zeta. \end{aligned}$$

Then, using $|J(s)| + |J'(s)| + |J''(s)| + |J'''(s)| \lesssim 1$ and $|J'(s)| \lesssim |s|$, we have

$$|H(\rho, \tau)| \leq C \frac{1 + \rho^2}{1 + \tau^2}, \quad |\partial_\rho H(\rho, \tau)| \leq C \frac{|\rho| + \rho^2}{1 + \tau^2}. \quad (3.11)$$

Moreover, since $H(\rho, \tau) = \mathcal{F}[h(\rho, \cdot)\mathbf{1}_{[-1,1]}](\tau)$, it holds

$$\int_{\mathbb{R}} H(\rho, \tau) d\tau = 2\pi h(\rho, 0) = 2\pi J(\rho).$$

For the estimate of $I^-(\tau, y)$, the change of the variable $\tau = t(ry_3 - 1)$ yields

$$I^-(t, y) = \frac{1}{ty_3} \int_{-t}^{\infty} g\left(\frac{\tau + t}{ty_3}\right) H\left(\frac{(\tau + t)|y_h|}{y_3}, \tau\right) d\tau,$$

where $g(r) := r^3 e^{-r^2}$. Then, we see that

$$\begin{aligned} I^-(t, y) &= \frac{2\pi}{ty_3^4} \exp\left(-\frac{1}{y_3^2}\right) J\left(\frac{t|y_h|}{y_3}\right) \\ &= I^-(t, y) - \frac{1}{ty_3} g\left(\frac{1}{y_3}\right) \int_{\mathbb{R}} H\left(\frac{t|y_h|}{y_3}, \tau\right) d\tau \\ &= E_1(t, y) + E_2(t, y) + E_3(t, y) + E_4(t, y), \end{aligned}$$

where

$$\begin{aligned} E_1(t, y) &:= \frac{1}{ty_3} g\left(\frac{1}{y_3}\right) \int_{-\infty}^{-t} H\left(\frac{t|y_h|}{y_3}, \tau\right) d\tau, \\ E_2(t, y) &:= \frac{1}{ty_3} \int_{-t}^t \left(g\left(\frac{\tau + t}{ty_3}\right) - g\left(\frac{1}{y_3}\right)\right) H\left(\frac{t|y_h|}{y_3}, \tau\right) d\tau, \\ E_3(t, y) &:= \frac{1}{ty_3} \int_{-t}^t g\left(\frac{\tau + t}{ty_3}\right) \left(H\left(\frac{(\tau + t)|y_h|}{y_3}, \tau\right) - H\left(\frac{t|y_h|}{y_3}, \tau\right)\right) d\tau \end{aligned}$$

$$E_4(t, y) := \frac{1}{ty_3} \int_t^\infty g\left(\frac{\tau+t}{ty_3}\right) H\left(\frac{(\tau+t)|y_h|}{y_3}, \tau\right) - g\left(\frac{1}{y_3}\right) H\left(\frac{t|y_h|}{y_3}, \tau\right) d\tau,$$

Making use of (3.11), we have

$$\begin{aligned} \sum_{m=1,4} \sup_{1 \leq y_3 \leq 2} |E_m(t, y)| &\leq C \frac{1+t^2|y_h|^2}{t} \int_{|\tau| \geq t} \frac{1}{1+\tau^2} d\tau \\ &\leq C \frac{1+t^2|y_h|^2}{t^2} \\ \sup_{1 \leq y_3 \leq 2} |E_2(t, y)| &\leq \int_{-t}^t \frac{|\tau|}{t} \left| H\left(\frac{t|y_h|}{y_3}, \tau\right) \right| d\tau \\ &\leq C \frac{1+t^2|y_h|^2}{t^2} \int_{-t}^t \frac{|\tau|}{1+\tau^2} d\tau \\ &\leq C(1+t^2|y_h|^2) \frac{\log t}{t^2}, \\ \sup_{1 \leq y_3 \leq 2} |E_3(t, y)| &\leq \int_{-t}^t \frac{|\tau||y_h|}{t} \sup_{0 \leq \rho \leq 2t|y_h|/y_3} |\partial_\rho H(\rho, \tau)| d\tau \\ &\leq C \int_{-t}^t \frac{|\tau||y_h|}{t} \frac{t|y_h|(1+t|y_h|)}{1+\tau^2} d\tau \\ &\leq C|y_h|^2(1+t|y_h|) \log t. \end{aligned}$$

Thus, we obtain

$$\begin{aligned} \sup_{1 \leq y_3 \leq 2} \left| I^-(t, y) - \frac{2\pi}{ty_3^4} \exp\left(-\frac{1}{y_3^2}\right) J\left(\frac{t|y_h|}{y_3}\right) \right| \\ \leq C(1+t^2|y_h|^2) \frac{\log t}{t^2} + C|y_h|^2(1+t|y_h|) \log t. \end{aligned}$$

For the estimate of $I^+(t, y)$, we have by (3.11) that

$$\begin{aligned} |I^+(t, y)| &\leq \int_0^\infty r^3 e^{-r^2} |H(tr|y_h|, t(ry_3+1))| dr \\ &\leq C \int_0^\infty r^3 e^{-r^2} \frac{1+t^2r^2|y_h|^2}{1+t^2(ry_3+1)^2} dr \\ &\leq \frac{C}{t^2} \int_0^\infty r^3 e^{-r^2} (1+t^2r^2|y_h|^2) dr = C \frac{1+t^2|y_h|^2}{t^2}. \end{aligned}$$

Let $K_{t,\delta} := \{y \in \mathbb{R}^3; |y_h| \leq \delta t^{-1}, 1 \leq y_3 \leq 2\}$ for $0 < \delta \leq 1$. Then, we have

$$\begin{aligned} \|U(t, \cdot)\|_{L^p} &\geq \|U_3(t, \cdot)\|_{L^p(K_{t,\delta})} \\ &\geq c \left\| \frac{2\pi}{ty_3^4} \exp\left(-\frac{1}{y_3^2}\right) J\left(\frac{t|y_h|}{y_3}\right) \right\|_{L^p(K_{t,\delta})} \\ &\quad - C \left\| I^-(t, y) - \frac{2\pi}{ty_3^4} \exp\left(-\frac{1}{y_3^2}\right) J\left(\frac{t|y_h|}{y_3}\right) \right\|_{L^\infty(K_{t,\delta})} |K_{t,\delta}|^{\frac{1}{p}} \\ &\quad - C \|I^+(t, \cdot)\|_{L^\infty(K_{t,\delta})} |K_{t,\delta}|^{\frac{1}{p}} \\ &\geq c \left(\inf_{|r| \leq C\delta} |J(r)| - C \frac{\log t}{t} - \frac{C}{t} \right) \frac{|K_{t,\delta}|^{\frac{1}{p}}}{t} \\ &\geq ct^{-1-\frac{2}{p}} \end{aligned}$$

for $t \geq e$ and sufficiently small δ .

Step 3. The L^4 estimate. Let $K_t := \{y \in \mathbb{R}^3 ; |y_h| \leq t^{-\frac{7}{8}}, 1 \leq y_3 \leq 2\}$. By the similar calculation as the last estimate in the above step, we have

$$\begin{aligned}
\|U(t, \cdot)\|_{L^4} &\geq \|U_3(t, \cdot)\|_{L^4(K_t)} \\
&\geq c \left\| \frac{2\pi}{ty_3^4} \exp\left(-\frac{1}{y_3^2}\right) J\left(\frac{t|y_h|}{y_3}\right) \right\|_{L^4(K_t)} \\
&\quad - C \left\| I^-(t, y) - \frac{2\pi}{ty_3^4} \exp\left(-\frac{1}{y_3^2}\right) J\left(\frac{t|y_h|}{y_3}\right) \right\|_{L^\infty(K_t)} |K_t|^{\frac{1}{4}} \\
&\quad - C \|I^+(t, \cdot)\|_{L^\infty(K_t)} |K_t|^{\frac{1}{4}} \\
&\geq \frac{c}{t} \left\| J\left(\frac{t|y_h|}{y_3}\right) \right\|_{L^4(K_t)} - C \frac{\log t}{t^{\frac{13}{8}}} |K_t|^{\frac{1}{4}} - \frac{C}{t^{\frac{7}{4}}} |K_t|^{\frac{1}{4}} \\
&\geq \frac{c}{t} \left\| J\left(\frac{t|y_h|}{y_3}\right) \right\|_{L^4(K_t)} - C \frac{\log t}{t^{\frac{33}{16}}}.
\end{aligned}$$

Changing the horizontal variable to the polar coordinate, we estimate the leading term as

$$\begin{aligned}
\left\| J\left(\frac{t|y_h|}{y_3}\right) \right\|_{L^4(K_t)}^4 &= 2\pi \int_1^2 \int_0^{t^{-7/8}} \left| J\left(\frac{tr}{y_3}\right) \right|^4 r dr dy_3 \\
&= \frac{2\pi}{t^2} \int_1^2 y_3^2 \int_0^{t^{1/8}/y_3} |J(\rho)|^4 \rho d\rho dy_3 \\
&\geq \frac{c}{t^2} \int_1^{ct^{1/8}} |J(\rho)|^4 \rho d\rho.
\end{aligned}$$

Since J is the zeroth order Bessel function, its asymptotic expansion is given by

$$\left| J(\rho) - \left(\frac{2}{\pi\rho}\right)^{\frac{1}{2}} \cos\left(\rho - \frac{\pi}{4}\right) \right| \leq C\rho^{-\frac{3}{2}}$$

for $\rho \geq 1$; see [15, Section B.8]. Using this and $\cos^4 \theta = (3 + 4 \cos 2\theta + \cos 4\theta)/8$, we have

$$\begin{aligned}
\int_1^{ct^{1/8}} |J(\rho)|^4 \rho d\rho &\geq c \int_1^{ct^{1/8}} \cos^4\left(\rho - \frac{\pi}{4}\right) \frac{d\rho}{\rho} - C \int_1^\infty \rho^{-5} d\rho \\
&\geq c \int_1^{ct^{1/8}} \frac{d\rho}{\rho} \\
&\quad - C \left| \int_1^{ct^{1/8}} \frac{\sin(2\rho)}{\rho} d\rho \right| - C \left| \int_1^{ct^{1/8}} \frac{\cos(4\rho)}{\rho} d\rho \right| - C \\
&\geq c \log t
\end{aligned}$$

for sufficiently large t . Combining above estimates, we obtain

$$\|U(t, \cdot)\|_{L^4} \geq Ct^{-\frac{3}{2}} (\log t)^{\frac{1}{4}},$$

which completes the proof. $\square$

Remark 3.8. Improving the calculations in the above proof with $2 \leq p < 4$, one may prove the following asymptotic expansion of the linear solution.

$$\lim_{t \rightarrow \infty} t^{\frac{3}{2}(1-\frac{1}{p})+\frac{1}{2}} (1 + |\Omega|t)^{\frac{3}{2}(1-\frac{2}{p})} \|\mathcal{T}_\Omega(t)u_0 - \mathcal{U}_\Omega[u_0](t, \cdot)\|_{L^p} = 0 \quad \text{for } 2 \leq p < 4,$$

where

$$\mathcal{U}_\Omega[u_0](t, x) := \frac{(|\Omega|t)^{\frac{5}{2}}}{(2\pi)^{\frac{3}{2}}} \frac{x_3^2}{|x_h|^{\frac{1}{2}}|x|^{\frac{9}{2}}} \exp\left(-\Omega^2 t^3 \frac{x_3^2}{|x|^4}\right) \left\{ \sin\left(|\Omega|t \frac{|x_h|}{|x|} - \frac{\pi}{4}\right) I
\right.$$

$$+ \operatorname{sgn}(\Omega) \cos \left(|\Omega| t \frac{|x_h|}{|x|} - \frac{\pi}{4} \right) Q(x) \Big\} M[u_0],$$

where

$$Q(x) := \frac{1}{|x_h||x|} \begin{pmatrix} 0 & |x_h|^2 & x_2 x_3 \\ -|x_h|^2 & 0 & -x_1 x_3 \\ -x_2 x_3 & x_1 x_3 & 0 \end{pmatrix},$$

$$M[u_0](x) := \sum_{k=1}^2 \frac{x_k x_3^2}{|x_h||x|^3} \int_{\mathbb{R}^3} y_k u_0(y) dy - \frac{|x_h| x_3}{|x|^3} \int_{\mathbb{R}^3} y_3 u_0(y) dy,$$

provided that $|x|u_0(x) \in L^1(\mathbb{R}^3)$. We do not present the precise proof of this expansion since the asymptotic analysis is not our main purpose of this paper.

4. NONLINEAR ANALYSIS

In this section, we present the proofs of our main results. We first recall the global well-posedness.

Lemma 4.1. *For any $u_0 \in H^{\frac{1}{2}}(\mathbb{R}^3)$ with $\operatorname{div} u_0 = 0$, there exists a positive constant $\Omega_0 = \Omega_0(u_0)$ suitably large such that if $|\Omega| \geq \Omega_0$, the system (1.1) possesses a unique global solution u satisfying*

$$u \in C([0, \infty); H^{\frac{1}{2}}(\mathbb{R}^3)), \quad \nabla u \in L^2(0, \infty; H^{\frac{1}{2}}(\mathbb{R}^3))$$

and, for every $m \in \mathbb{N}$, it holds $\nabla^m u \in C((0, \infty); H^{\frac{1}{2}}(\mathbb{R}^3))$ with the estimate

$$\sup_{t>0} t^{\frac{m}{2}} \|\nabla^m u(t)\|_{H^{\frac{1}{2}}} < \infty.$$

Moreover if we additionally assume $u_0 \in L^1(\mathbb{R}^3)$, then there exists a positive constant $K_{0,m} = K_{0,m}(\|u_0\|_{L^1}, \|u_0\|_{\dot{H}^{\frac{1}{2}}})$ such that

$$\|\nabla^m u(t)\|_{L^2} \leq K_{0,m} (1+t)^{-\frac{3}{4}} t^{-\frac{m}{2}}$$

for all $t > 0$. Furthermore, for $0 < \varepsilon \leq 1$, it holds $K_{0,m} = O_m(\varepsilon)$ if $\|u_0\|_{L^1 \cap \dot{H}^{\frac{1}{2}}} \leq \varepsilon$.

We omit the proof of this lemma since it is obtained by the global well-posedness theory of [19] and the Fourier splitting method established in [27].

Now we focus on the decay estimates of the nonlinear term given by

$$\mathcal{N}_\Omega[u](t) := \int_0^t \mathcal{T}_\Omega(t-\tau) \mathbb{P} \operatorname{div}(u \otimes u)(\tau) d\tau.$$

Proposition 4.2. *Let $u_0 \in \dot{H}^{\frac{1}{2}}(\mathbb{R}^3) \cap L^1(\mathbb{R}^3)$ with $\operatorname{div} u_0 = 0$, and let u be the associated global solution to (1.1) with $|\Omega| \geq \Omega_0$, constructed in Lemma 4.1. Then, for every $2 \leq p < \infty$ and $m \in \mathbb{N} \cup \{0\}$, there exists a positive constant $K_{p,m} = K_{p,m}(\|u_0\|_{L^1}, \|u_0\|_{\dot{H}^{\frac{1}{2}}})$, independent of Ω , such that*

$$\|\nabla^m \mathcal{N}_\Omega[u](t)\|_{L^p} \leq K_{p,m} t^{-\frac{3}{2}(1-\frac{1}{p})-\frac{m+1}{2}} \mathcal{D}_p(|\Omega|t), \quad (4.1)$$

$$\|\nabla^m \mathcal{N}_\Omega[u](t)\|_{L^{4,\infty}} \leq K_{4,m} t^{-\frac{3}{2}(1-\frac{1}{4})-\frac{m+1}{2}} (1+|\Omega|t)^{-\frac{3}{4}} \quad (4.2)$$

for all $t > 0$. Moreover, for the case of $p = \infty$, there exist a positive constant $K_{\infty,m} = K_{\infty,m}(\|u_0\|_{L^1}, \|u_0\|_{\dot{H}^{\frac{1}{2}}})$ and an absolute positive constant C such that

$$\begin{aligned} \|\nabla^m \mathcal{N}_\Omega[u](t)\|_{L^\infty} &\leq K_{\infty,m} t^{-\frac{3}{2}-\frac{m+1}{2}} \mathcal{D}_\infty(|\Omega|t) \\ &\quad + C t^{-\frac{5}{2}-\frac{m}{2}} \mathcal{D}_4(|\Omega|t)^2 \|u\|_{X_{\Omega,4}^0(t)} \|u\|_{X_{\Omega,4}^{m+1}(t)}^{\frac{1}{2}} \|u\|_{X_{\Omega,4}^{m+2}(t)}^{\frac{1}{2}} \end{aligned}$$

for all $t > 0$, where

$$\|u\|_{X_{\Omega,p}^m(t)} := \sup_{\frac{t}{2} \leq \tau \leq t} \tau^{\frac{3}{2}(1-\frac{1}{p})+\frac{m}{2}} \mathcal{D}_p(|\Omega|\tau)^{-1} \|\nabla^m u(\tau)\|_{L^p}.$$

Furthermore, it holds $K_{p,m} = O_{p,m}(\varepsilon^2)$ for $2 \leq p \leq \infty$, provided that $\|u_0\|_{L^1 \cap \dot{H}^{\frac{1}{2}}} \leq \varepsilon$ with $0 < \varepsilon \leq 1$.

To show this proposition, we use the following elementary estimate.

Lemma 4.3. *For $2 \leq p \leq \infty$, there exists a positive constant $C = C(p)$ such that*

$$\int_0^t \mathcal{D}_p(|\Omega|s) ds \leq \begin{cases} Ct \mathcal{D}_p(|\Omega|t) & (2 \leq p < \infty), \\ C|\Omega|^{-1} \log(e + |\Omega|t) & (p = \infty) \end{cases}$$

for $t > 0$ and $\Omega \in \mathbb{R} \setminus \{0\}$.

Proof. The case of $p = 2$ is trivial. For the case of $p = \infty$, we see that

$$\begin{aligned} \int_0^t (1 + |\Omega|\tau)^{-1} d\tau &= |\Omega|^{-1} \int_0^{|\Omega|t} (1 + s)^{-1} ds \\ &= |\Omega|^{-1} \log(1 + |\Omega|t) \leq |\Omega|^{-1} \log(e + |\Omega|t). \end{aligned}$$

Next, we focus on the case $2 < p < \infty$. In case of $p \neq 4$, we use the estimate

$$\int_0^t (1 + s)^{-\gamma} ds = \frac{(1 + t)^{1-\gamma} - 1}{1 - \gamma} \leq \frac{t(1 + t)^{-\gamma}}{1 - \gamma}$$

for $0 < \gamma < 1$ and $t > 0$. Let

$$\gamma_p := \begin{cases} \frac{3}{2}(1 - \frac{2}{p}) & (2 < p < 4), \\ 1 - \frac{1}{p} & (4 < p < \infty). \end{cases}$$

Then, we have for $p \neq 4$ that

$$\begin{aligned} \int_0^t \mathcal{D}_p(|\Omega|\tau) d\tau &= |\Omega|^{-1} \int_0^{|\Omega|t} (1 + s)^{-\gamma_p} ds \\ &\leq \frac{t(1 + |\Omega|t)^{-\gamma_p}}{1 - \gamma_p} = \frac{1}{1 - \gamma_p} t \mathcal{D}_p(|\Omega|t). \end{aligned}$$

For the case of $p = 4$, we have

$$\begin{aligned} \int_0^t \mathcal{D}_4(|\Omega|\tau) d\tau &= |\Omega|^{-1} \int_0^{|\Omega|t} (1 + s)^{-\frac{3}{4}} (\log(e + s))^{\frac{1}{4}} ds \\ &\leq |\Omega|^{-1} \int_0^{|\Omega|t} (1 + s)^{-\frac{3}{4}} ds (\log(e + |\Omega|t))^{\frac{1}{4}} \\ &\leq 4t(1 + |\Omega|t)^{-\frac{3}{4}} (\log(e + |\Omega|t))^{\frac{1}{4}} = 4t \mathcal{D}_4(|\Omega|t), \end{aligned}$$

which completes the proof. □

Proof of Proposition 4.2. Let us decompose the nonlinear term as

$$\begin{aligned} \mathcal{N}_\Omega[u](t) &= \underline{\mathcal{N}}_\Omega[u](t) + \overline{\mathcal{N}}_\Omega[u](t), \\ \underline{\mathcal{N}}_\Omega[u](t) &:= \int_0^{\frac{t}{2}} \mathcal{T}_\Omega(t - \tau) \mathbb{P} \operatorname{div}(u \otimes u)(\tau) d\tau, \\ \overline{\mathcal{N}}_\Omega[u](t) &:= \int_{\frac{t}{2}}^t \mathcal{T}_\Omega(t - \tau) \mathbb{P} \operatorname{div}(u \otimes u)(\tau) d\tau. \end{aligned}$$

We see by Theorem 3.2 and Lemma 4.1 that for every $2 \leq p \leq \infty$,

$$\begin{aligned}
\|\nabla^m \mathcal{N}_\Omega[u](t)\|_{L^p} &\leq C \int_0^t (t-\tau)^{-\frac{3}{2}(1-\frac{1}{p})-\frac{m+1}{2}} \mathcal{D}_p(|\Omega|(t-\tau)) \|u(\tau) \otimes u(\tau)\|_{L^1} d\tau \\
&\leq Ct^{-\frac{3}{2}(1-\frac{1}{p})-\frac{m+1}{2}} \mathcal{D}_p(|\Omega|t) \int_0^\infty \|u(\tau)\|_{L^2}^2 d\tau \\
&\leq Ct^{-\frac{3}{2}(1-\frac{1}{p})-\frac{m+1}{2}} \mathcal{D}_p(|\Omega|t) \int_0^\infty (1+\tau)^{-\frac{3}{2}} d\tau \\
&= Ct^{-\frac{3}{2}(1-\frac{1}{p})-\frac{m+1}{2}} \mathcal{D}_p(|\Omega|t).
\end{aligned} \tag{4.3}$$

Step 1. The L^p estimate of $\overline{\mathcal{N}}_\Omega[u]$ for $2 \leq p < \infty$. Based on Lemma 4.1, Proposition 3.3, the product estimate from Lemma 2.3 and the interpolation inequality from Lemma 2.2, we infer that

$$\begin{aligned}
\|\nabla^m \overline{\mathcal{N}}_\Omega[u]\|_{L^p} &\leq C \int_{\frac{t}{2}}^t \mathcal{D}_p(\Omega(t-\tau)) \|u(\tau) \otimes u(\tau)\|_{\dot{B}_{1,1}^{m+1+3(1-\frac{1}{p})}} d\tau \\
&\leq C \int_{\frac{t}{2}}^t \mathcal{D}_p(\Omega(t-\tau)) \|u(\tau)\|_{L^2} \|u(\tau)\|_{\dot{B}_{2,1}^{m+1+3(1-\frac{1}{p})}} d\tau \\
&\leq C \int_{\frac{t}{2}}^t \mathcal{D}_p(\Omega(t-\tau)) \|u(\tau)\|_{L^2} \|\nabla^{m+1} u(\tau)\|_{L^2}^{\frac{1}{p}} \|\nabla^{m+4} u(\tau)\|_{L^2}^{1-\frac{1}{p}} d\tau \\
&\leq C \int_{\frac{t}{2}}^t \mathcal{D}_p(\Omega(t-\tau)) (1+\tau)^{-\frac{3}{2}} \tau^{-\frac{m+1}{2}-\frac{3}{2}(1-\frac{1}{p})} d\tau \\
&\leq C(1+t)^{-\frac{3}{2}t-\frac{m+1}{2}-\frac{3}{2}(1-\frac{1}{p})} \int_0^{\frac{t}{2}} \mathcal{D}_p(|\Omega|s) ds \\
&\leq C(1+t)^{-\frac{3}{2}t-\frac{m+1}{2}-\frac{3}{2}(1-\frac{1}{p})} \mathcal{D}_p(|\Omega|t).
\end{aligned} \tag{4.4}$$

Combining (4.3) and (4.4), we complete the proof of (4.1) with $2 \leq p < \infty$. We emphasize that the same argument, due to the second assertion of Proposition 3.3, leads to

$$\|\nabla^m \overline{\mathcal{N}}_\Omega[u]\|_{L^{4,\infty}} \leq C(1+t)^{-\frac{3}{2}t-\frac{m+1}{2}-\frac{3}{2}(1-\frac{1}{4})} (1+|\Omega|t)^{-\frac{3}{4}}.$$

Similarly, we get

$$\|\nabla^m \mathcal{N}_\Omega[u]\|_{L^{4,\infty}} \leq Ct^{-\frac{3}{2}(1-\frac{1}{4})-\frac{m+1}{2}} (1+|\Omega|t)^{-\frac{3}{4}}.$$

The two estimates above verify (4.2).

Step 2. The L^∞ estimate of $\overline{\mathcal{N}}_\Omega[u]$. By Proposition 3.3 and Lemmas 2.1 and 2.3, it holds

$$\begin{aligned}
\|\nabla^m \overline{\mathcal{N}}_\Omega[u](t)\|_{L^\infty} &\leq C \sum_{\sigma \in \{\pm\}} \int_{\frac{t}{2}}^t \left\| e^{(t-\tau)\Delta} P_\sigma(D) \mathbb{P} \operatorname{div}(u(\tau) \otimes u(\tau)) \right\|_{\dot{B}_{2,1}^{m+\frac{3}{2}}} d\tau \\
&\leq C \int_{\frac{t}{2}}^t (t-\tau)^{-\frac{1}{2}} \|u(\tau) \otimes u(\tau)\|_{\dot{B}_{2,\infty}^{m+\frac{3}{2}}} d\tau \\
&\leq C \int_{\frac{t}{2}}^t (t-\tau)^{-\frac{1}{2}} \|u(\tau)\|_{L^4} \|u(\tau)\|_{\dot{B}_{4,\infty}^{m+\frac{3}{2}}} d\tau \\
&\leq C \int_{\frac{t}{2}}^t (t-\tau)^{-\frac{1}{2}} \|u(\tau)\|_{L^4} \|\nabla^{m+1} u(\tau)\|_{L^4}^{\frac{1}{2}} \|\nabla^{m+2} u(\tau)\|_{L^4}^{\frac{1}{2}} d\tau \\
&\leq C \int_{\frac{t}{2}}^t (t-\tau)^{-\frac{1}{2}} \tau^{-3-\frac{m}{2}} \mathcal{D}_4(|\Omega|\tau)^2 d\tau \|u\|_{X_{\Omega,4}^0(t)} \|u\|_{X_{\Omega,4}^{m+1}(t)}^{\frac{1}{2}} \|u\|_{X_{\Omega,4}^{m+2}(t)}^{\frac{1}{2}} \\
&\leq Ct^{-\frac{5}{2}-\frac{m}{2}} \mathcal{D}_4(\Omega t)^2 \|u\|_{X_{\Omega,4}^0(t)} \|u\|_{X_{\Omega,4}^{m+1}(t)}^{\frac{1}{2}} \|u\|_{X_{\Omega,4}^{m+2}(t)}^{\frac{1}{2}}.
\end{aligned}$$

Hence, gathering the above estimates, we complete the proof. $\square$

Now, we are in a position to prove our main results.

Proof of Theorems 1.1 and 1.2. Let us first prove the decay rate of small order (1.7). It follows from Theorem 3.2 and Proposition 4.2 that for $2 \leq p < \infty$,

$$\begin{aligned} \|\nabla^m u(t)\|_{L^p} &\leq \|\nabla^m \mathcal{T}_\Omega(t)u_0\|_{L^p} + \|\nabla^m \mathcal{N}_\Omega[u](t)\|_{L^p} \\ &\leq \|\nabla^m \mathcal{T}_\Omega(t)u_0\|_{L^p} + K_{p,m} t^{-\frac{3}{2}(1-\frac{1}{p})-\frac{m+1}{2}} \mathcal{D}_p(|\Omega|t) \\ &= o\left(t^{-\frac{3}{2}(1-\frac{1}{p})-\frac{m}{2}} \mathcal{D}_p(|\Omega|t)\right) \end{aligned} \quad (4.5)$$

as $t \rightarrow \infty$. On the L^∞ case, we see by Theorem 3.2 and Proposition 4.2 that

$$\begin{aligned} \|\nabla^m u(t)\|_{L^p} &\leq \|\nabla^m \mathcal{T}_\Omega(t)u_0\|_{L^\infty} + \|\nabla^m \mathcal{N}_\Omega[u](t)\|_{L^\infty} \\ &\leq \|\nabla^m \mathcal{T}_\Omega(t)u_0\|_{L^\infty} + K_{\infty,m} t^{-\frac{3}{2}-\frac{m+1}{2}} \mathcal{D}_\infty(|\Omega|t) \\ &\quad + Ct^{-\frac{5}{2}-\frac{m}{2}} \mathcal{D}_4(|\Omega|t)^2 \|u\|_{X_{\Omega,4}^0(t)} \|u\|_{X_{\Omega,4}^{m+1}(t)}^{\frac{1}{2}} \|u\|_{X_{\Omega,4}^{m+2}(t)}^{\frac{1}{2}} \\ &= o\left(t^{-\frac{3}{2}-\frac{m+1}{2}} \mathcal{D}_\infty(|\Omega|t)\right) \end{aligned}$$

as $t \rightarrow \infty$, where we have used $\|u\|_{X_{\Omega,4}^0(t)} \|u\|_{X_{\Omega,4}^{m+1}(t)}^{\frac{1}{2}} \|u\|_{X_{\Omega,4}^{m+2}(t)}^{\frac{1}{2}} = o(1)$ ($t \rightarrow \infty$) due to (4.5) with $p = 4$.

Next, we focus on the enhanced decay estimates (1.8), where we additionally assume that $|x|u_0(x) \in L^1(\mathbb{R}^3)$. In case of $2 \leq p < \infty$, it follows from Theorem 3.2 and (4.1) that

$$\begin{aligned} \|\nabla^m u(t)\|_{L^p} &\leq \|\nabla^m \mathcal{T}_\Omega(t)u_0\|_{L^p} + \|\nabla^m \mathcal{N}_\Omega[u](t)\|_{L^p} \\ &\leq Ct^{-\frac{3}{2}(1-\frac{1}{p})-\frac{m+1}{2}} \mathcal{D}_p(|\Omega|t) \| |x|u_0(x) \|_{L^1(\mathbb{R}_x^3)} \\ &\quad + K_{p,m} t^{-\frac{3}{2}(1-\frac{1}{p})-\frac{m+1}{2}} \mathcal{D}_p(|\Omega|t) \\ &\leq Ct^{-\frac{3}{2}(1-\frac{1}{p})-\frac{m+1}{2}} \mathcal{D}_p(|\Omega|t) \end{aligned} \quad (4.6)$$

for all $t > 0$. It holds by Theorem 3.2, Proposition 3.6, and (4.2) that

$$\begin{aligned} \|\nabla^m u(t)\|_{L^{4,\infty}} &\leq C \|\nabla^m \mathcal{T}_\Omega(t)u_0\|_{L^{4,\infty}} + C \|\nabla^m \mathcal{N}_\Omega[u](t)\|_{L^{4,\infty}} \\ &\leq Ct^{-\frac{3}{2}(1-\frac{1}{4})-\frac{m+1}{2}} (1 + |\Omega|t)^{-\frac{3}{4}} \| |x|u_0(x) \|_{L^1(\mathbb{R}_x^3)} \\ &\quad + K_{4,m} t^{-\frac{3}{2}(1-\frac{1}{4})-\frac{m+1}{2}} (1 + |\Omega|t)^{-\frac{3}{4}} \\ &\leq Ct^{-\frac{3}{2}(1-\frac{1}{4})-\frac{m+1}{2}} (1 + |\Omega|t)^{-\frac{3}{4}} \end{aligned}$$

for all $t > 0$. Finally, we consider the decay estimates in L^∞ . We have by Proposition 4.2 that

$$\begin{aligned} \|\nabla^m u(t)\|_{L^\infty} &\leq \|\nabla^m \mathcal{T}_\Omega(t)u_0\|_{L^\infty} + \|\nabla^m \mathcal{N}_\Omega[u](t)\|_{L^\infty} \\ &\leq Ct^{-\frac{3}{2}-\frac{m+1}{2}} \mathcal{D}_\infty(|\Omega|t) \| |x|u_0(x) \|_{L^1(\mathbb{R}_x^3)} \\ &\quad + K_{\infty,m} t^{-\frac{3}{2}-\frac{m+1}{2}} \mathcal{D}_\infty(|\Omega|t) \\ &\quad + Ct^{-\frac{5}{2}-\frac{m}{2}} \mathcal{D}_4(|\Omega|t)^2 \|u\|_{X_{\Omega,4}^0(t)} \|u\|_{X_{\Omega,4}^{m+1}(t)}^{\frac{1}{2}} \|u\|_{X_{\Omega,4}^{m+2}(t)}^{\frac{1}{2}} \\ &\leq Ct^{-\frac{3}{2}-\frac{m+1}{2}} \mathcal{D}_\infty(|\Omega|t) \end{aligned}$$

for all $t \geq 1$, where we have used

$$\sup_{t>0} \left(\|u\|_{X_{\Omega,4}^0(t)} \|u\|_{X_{\Omega,4}^{m+1}(t)}^{\frac{1}{2}} \|u\|_{X_{\Omega,4}^{m+2}(t)}^{\frac{1}{2}} \right) \leq C,$$

which is implied by (4.6). Thus, we complete the proof. $\square$

Proof of Theorem 1.4. Let $0 < \varepsilon \leq 1$ and

$$u_0(x) := \varepsilon e^{-\frac{|x|^2}{4}} x^\perp.$$

Then, we see that $\|u_0\|_{L^1 \cap \dot{H}^{\frac{1}{2}}} \leq C\varepsilon$. Choosing ε sufficiently small, the Fujita–Kato principle [14] implies that u_0 generates a global small smooth solution u to (1.1) with $\Omega = 1$. From Theorem 3.7 and the decay estimates of nonlinear term $\mathcal{N}_\Omega[u]$ obtained in the proof of Theorem 1.1, it holds

$$\begin{aligned} \|u(t)\|_{L^p} &\geq \|\mathcal{T}_\Omega u_0\|_{L^p} - \|\mathcal{N}_\Omega[u](t)\|_{L^p} \\ &\geq c\varepsilon t^{-\frac{3}{2}(1-\frac{1}{p})-\frac{1}{2}} \mathcal{D}_p(t) - C\varepsilon^2 t^{-\frac{3}{2}(1-\frac{1}{p})-\frac{1}{2}} \mathcal{D}_p(t), \\ \|u(t)\|_{L^{4,\infty}} &\geq \|\mathcal{T}_\Omega u_0\|_{L^{4,\infty}} - \|\mathcal{N}_\Omega[u](t)\|_{L^{4,\infty}} \\ &\geq c\varepsilon t^{-\frac{3}{2}(1-\frac{1}{4})-\frac{1}{2}} (1+t)^{-\frac{3}{4}} - C\varepsilon^2 t^{-\frac{3}{2}(1-\frac{1}{4})-\frac{1}{2}} (1+t)^{-\frac{3}{4}} \end{aligned}$$

for all sufficiently large t . Choosing ε sufficiently small, we complete the proof. $\square$

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